



School of AI & Data for Business & Society

Paris Region · Nice Côte d'Azur

Grande Ecole Programme – PGE5

Specialisation: Artificial Intelligence & Data Science

FINAL YEAR PROJECT REPORT (PFE)

Where RAG Breaks

Citation-Grounded Question Answering and Escalation over EU Aviation Regulations

Student:	Mahamat Saleh Samir Adam
Host Company:	Direction générale de l'Aviation civile (DGAC) – DSAC / MEAS / ERAS
Industry Supervisor:	Romain BUFFRY, Chef de Pôle — “Évaluations des Risques et Analyses de sécurité” (MEAS/ERAS)
Academic Supervisor:	Antoun YAACOUB (aivancity)
Academic Year:	2025–2026
Submission Date:	31 August 2026

Defence date: 23 September 2026

Acknowledgements

First and above all, I thank Allah, for everything.

I thank Antoun Yaacoub, my academic supervisor at aivancity, for the methodological direction that turned a system comparison into a research question, and for insisting that the evaluation instrument be validated before its measurements were believed. The central argument of this work exists because of that insistence.

I thank Yoni Malka, who supervised my apprenticeship at the Direction générale de l'Aviation civile as head of the *Évaluations des Risques et Analyses de sécurité* division, and my colleagues in MEAS/ERAS for the professional context in which the questions this thesis asks became worth asking.

I thank the four aviation professionals who reviewed the benchmark independently and who are identified here only by pseudonym. Their disagreement was more useful to this work than their agreement would have been, and Chapter 5 is built on it.

I thank my parents, Samir and Sarra, for everything they gave up so that this page could be written. To my father: you sent me abroad to study and carried the cost of it, in every sense, for all these years, and you never once let me feel it as a burden. To my mother and my sisters: you carried me through the stretches when I could not carry myself, and you did it without being asked. I thank my brothers and sisters for their steady support throughout. I am grateful in a way this paragraph cannot hold.

And I thank my friends, who kept me company through a long year and never let it become a lonely one.

Abstract

Large language models answer regulatory questions fluently while citing the wrong provision, quoting a rule that omits its own exception, or presenting an unsupported conclusion. In aviation such failures are consequential: an answer must be traceable to authoritative, versioned legal evidence. This project studies citation-grounded question answering over Regulation (EU) No 376/2014 on occurrence reporting, reframing an initial comparison of few-shot prompting, retrieval-augmented generation and parameter-efficient fine-tuning into a failure-analysis question: where does retrieval-grounded question answering break down, and which condition observable at runtime justifies escalating instead of answering?

The evaluation instrument was validated before its measurements were believed. Four aviation professionals independently reviewed a bounded sample of the benchmark; they agreed on the verdict for fewer than half of the records (Fleiss' $\kappa = 0.250$), which is reported as a finding rather than resolved away. The reviewed benchmark of 98 evidence-linked questions was then frozen under a pre-registered rule.

Retrieval is not where the system fails: hybrid retrieval with cross-encoder reranking places acceptable evidence in the top five results for 89.3% of held-out questions. The failure is citation precision. Under an evaluator that resolves references against the corpus rather than matching strings, the retrieval-augmented baseline identifies the correct article for 62.5% of questions but reproduces the complete citation, sub-paragraph included, for 44.3%.

Two results shape the resulting architecture. Deterministic quote verification and evidence-selection errors did not co-occur once, so both checks are required. And across 140 generated answers the model never once used its explicit, cost-free abstention channel — including on answers that fabricated a source identifier. Abstention cannot be delegated to the generator; it must be imposed from outside, on evidence the system can check. The escalation policy is specified accordingly, with its retrieval-uncertainty threshold left explicitly uncalibrated and the sample size that calibration would require stated.

Keywords: *retrieval-augmented generation, legal question answering, aviation regulation, hybrid retrieval, reranking, grounded generation, citation verification, abstention, selective prediction.*

Résumé (French Summary)

Les grands modèles de langage répondent aux questions réglementaires de manière convaincante tout en citant la mauvaise disposition, en amputant une règle de son exception ou en avançant une conclusion non étayée. En aéronautique, une réponse doit être rattachée à une source juridique officielle, versionnée et précisément citée. Ce projet étudie la question-réponse fondée sur les preuves appliquée au règlement (UE) n° 376/2014, et transforme une comparaison de systèmes en analyse des modes de défaillance : où le système échoue-t-il, et quelle condition observable à l'exécution justifie une escalade plutôt qu'une réponse ?

L'instrument de mesure a été validé avant d'être cru. Quatre professionnels de l'aéronautique ont examiné indépendamment un échantillon borné du benchmark ; ils ne s'accordent sur le verdict que pour moins de la moitié des enregistrements (κ de Fleiss = 0,250). Ce désaccord est rapporté comme un résultat. Le benchmark revu, 98 questions liées à leurs preuves, a été gelé selon une règle préenregistrée.

La recherche d'information n'est pas le point de rupture : la recherche hybride avec reranker cross-encoder place une preuve acceptable dans les cinq premiers résultats pour 89,3 % des questions du jeu de test. La défaillance porte sur la précision des citations : réévaluée avec un évaluateur qui résout les références contre le corpus, la base RAG identifie le bon article pour 62,5 % des questions mais ne reproduit la citation complète, sous-paragraphe inclus, que pour 44,3 %.

Deux résultats déterminent l'architecture. Les erreurs de vérification des citations et de sélection des preuves ne coïncident jamais : les deux contrôles sont nécessaires. Et sur 140 réponses générées, le modèle n'a jamais utilisé le canal d'abstention explicite et sans coût qui lui était offert, y compris pour une réponse citant un identifiant inventé. L'abstention doit être imposée de l'extérieur, sur des éléments vérifiables. La politique d'escalade est spécifiée en conséquence, son seuil laissé explicitement non calibré.

Mots-clés : *RAG, question-réponse juridique, réglementation aéronautique, recherche hybride, reranking, vérification des citations, abstention, prédiction sélective.*

Declaration

I declare that this final year project report is my own work, carried out under the supervision named on the cover page, and that it has not been submitted for any other qualification.

All sources consulted are cited in the body of the text and listed in the bibliography. Every direct quotation from a legal instrument, guidance document or publication is identified as such.

Generative artificial intelligence tools were used during this project. In accordance with the aivancity rules on the use of generative AI, every use is disclosed in Appendix B, which records the tool, the period of use, the task it assisted with, what was produced, and what was verified by me before inclusion. The reasoning, the experimental decisions, the adjudication of the reviewed benchmark, and the conclusions drawn are my own, and I take responsibility for the whole of this document.

The independent reviewers who assessed the benchmark are referred to by pseudonym. Their identities are known to me and are not disclosed here.

No internal document, dataset or occurrence report of the host organisation was used at any stage of this work.

Mahamat Saleh Samir Adam

Date: 31 August 2026

Contents

Acknowledgements	I
Abstract	II
Résumé (French Summary)	III
Declaration	IV
List of Figures	IX
List of Tables	XII
List of Abbreviations and Acronyms	XIII
1 Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement	2
1.3 Methodological Stance	3
1.4 Objectives and Contributions	4
1.5 Scope and Delimitations	5
1.6 Report Structure	6
2 State of the Art	7
2.1 Scope and Organisation	7
2.2 Retrieval-Augmented Generation	7
2.3 Retrieval or Parametric Adaptation?	9
2.4 Sparse, Dense, and Hybrid Retrieval	9
2.5 Evidence Granularity and Document Hierarchy	11
2.6 Cross-Encoder Reranking	12
2.7 Legal and Regulatory Language Processing	13
2.8 Attribution, Citation, and Verifiability	14
2.9 Selective Answering, Calibration, and Escalation	15
2.10 Evaluating Retrieval and Generation	16
2.11 Benchmark Validity and Reviewer Agreement	17
2.12 Agentic and Multi-Step Retrieval	17
2.13 Positioning of This Work	18
3 Project Context and Scope	20

3.1	Host Organisation and Team	20
3.2	The Regulatory Domain	21
3.3	Intended Users and Use	22
3.4	Mission and Its Revision	22
3.5	Data and Provenance	23
3.6	Constraints	24
3.7	Infrastructure	24
3.8	Governance and Traceability	25
3.9	Deliverable Schedule	25
3.10	Risks and Mitigations	26
4	Methodology	28
4.1	Research Design	28
4.1.1	Design Principles	28
4.1.2	Experiment Ladder	30
4.1.3	Unit of Analysis	31
4.2	Corpus Construction	31
4.2.1	Defects in the Historical Corpus	31
4.2.2	Source Snapshot and Provenance	31
4.2.3	Legal Identity	32
4.2.4	Parent-Child Representation	32
4.2.5	Deterministic Validation	33
4.3	Benchmark Construction	34
4.3.1	Legacy Lineage	34
4.3.2	Evidence Audit of the Shared Evaluation Set	34
4.3.3	Independent Multi-Reviewer Validation	35
4.3.4	Parent-Grouped Partitioning	36
4.4	Retrieval Architecture	37
4.4.1	Dense Retrieval	37
4.4.2	Lexical Retrieval	37
4.4.3	Rank Fusion	38
4.4.4	Cross-Encoder Reranking	38
4.5	Context Construction and the Generation Contract	39
4.5.1	Context Selection	39
4.5.2	Answer Schema	40
4.5.3	Prompt Contract	41
4.6	Verification and Selective Answering	41
4.6.1	Structural Verification	41
4.6.2	The Limits of Structural Verification	42
4.6.3	Selective Answering	43

4.7	Escalation Policy and Its Calibration	43
4.8	Evaluation Measures	44
4.8.1	Retrieval Measures	44
4.8.2	Generation Measures	44
4.8.3	Statistical Treatment	45
4.9	Threats to Validity	45
4.10	Reproducibility and Audit Trail	45
4.11	Ethical and Professional Considerations	46
5	Results and Analysis	48
5.1	Overview	48
5.2	The Corpus and the Frozen Benchmark	50
5.2.1	Structural corpus	50
5.2.2	The multi-reviewer validity gate	50
5.2.3	What the review changed	51
5.3	Retrieval	52
5.3.1	Dense embedding comparison	52
5.3.2	Lexical, hybrid and reranked retrieval	53
5.3.3	Effect of the review on measured performance	54
5.3.4	Rejected alternatives	55
5.4	The Three-System Baseline, Re-scored	55
5.4.1	What the original measure counted	55
5.4.2	Scope	56
5.4.3	Results	56
5.5	Evidence-Constrained Generation	58
5.5.1	A schema-conformant answer is not a contract-valid answer	59
5.5.2	What counts as a verbatim quote	59
5.5.3	One prompt rule, measured	60
5.5.4	The failure the fix displaced	61
5.5.5	The enriched context variant	62
5.5.6	The generator never abstains	63
5.6	Escalation Calibration	63
5.6.1	The sample cannot support a threshold	63
5.6.2	How much data would be required	64
5.6.3	The two failure modes are independent	64
5.6.4	The policy as specified	65
5.7	Two Properties No Internal Metric Detects	67
5.7.1	Citation labels a reader cannot resolve	67
5.7.2	The benchmark cannot demonstrate an agent's benefit	67
5.8	Threats to Validity	68

6 Conclusion	70
6.1 Answer to the Research Question	70
6.1.1 Retrieval is not where this system fails	70
6.1.2 The failure is citation precision, and it is invisible to conventional measures	70
6.1.3 The escalation condition, and the honest limit on it	71
6.2 What the Project Established That It Did Not Set Out to Find	72
6.2.1 Verification and instruction are not substitutes	72
6.2.2 Abstention cannot be delegated to the generator	72
6.2.3 The two failure modes are independent	72
6.2.4 A benchmark's labels can be internally valid and externally unreadable	73
6.3 Contributions	73
6.4 Limitations	74
6.5 Future Work	75
6.6 Closing Remark	75
Bibliography	77
A Reproducibility and Cost	84
A.1 Execution Environments	84
A.2 Determinism and Caching	85
A.3 Hosted API Usage	85
A.4 Cost	86
A.5 Reproducing the Results	87
B Generative AI Use Disclosure	89
B.1 Three distinct kinds of use	89
B.2 Itemised record of research and writing assistance	90
B.3 What was not delegated	90
B.4 Provenance discipline	91
C Independent Benchmark Review Protocol	92
D Selected Source Code	93
D.1 Quote Grounding	93
D.2 Structural Verification	94
D.3 Citation Scoring	95

List of Figures

4.1	The evaluated pipeline, end to end	29
5.1	Held-out retrieval on the frozen benchmark	53
5.2	The two failure modes do not co-occur	65
5.3	The escalation policy and the status of each transition	66

List of Tables

2.1	Retrieval families and their principal trade-offs, as characterised by the cited work.	11
2.2	Escalation levels and the prior work that motivates each. The contribution of this thesis is the calibrated transition between levels, not the levels themselves.	19
3.1	Composition of the source material, from the validated corpus.	21
3.2	Principal project risks and the mitigation applied to each.	27
4.1	Experiment ladder. Each stage alters one component relative to the stage above it.	30
4.2	Stored fields. Parents carry fourteen fields, of which only cross-references are optional; all eleven child fields are required.	32
4.3	Validated E1 corpus composition, from the stored validation record. . .	33
4.4	Legacy parent-lineage audit over 300 historical records.	34
4.5	Candidate parent-grouped evaluation split, from the stored split summary.	36
4.6	Observed context statistics for the enriched variant, as built over the 98 frozen questions — the artefact that generation reads. The earlier build over the 99 pre-review questions gave 6.36 neighbours and 473.3 evidence words per question under the same selection rule.	40
4.7	Escalation levels and their trigger conditions. All thresholds are calibrated on development data only.	43
4.8	Principal threats to validity and the control applied to each.	46
5.1	Principal results of this chapter, with the section or table in which each is reported. Held-out figures come from a single run made after every component they evaluate had been frozen.	49
5.2	Independent review panel. Reviewers are pseudonymised; the mapping to identities is held separately and is not part of this work.	50
5.3	Inter-rater agreement on the overall verdict, computed over the 31 records with at least two raters.	51
5.4	Benchmark freeze outcome.	51

5.5	Development recall for the dense embedding comparison, reviewed labels. Proportion of questions retrieving at least one acceptable evidence child.	52
5.6	Held-out retrieval on the frozen reviewed benchmark ($n = 56$). Proportion of questions retrieving at least one acceptable evidence child. . . .	53
5.7	Held-out recall for hybrid retrieval with cross-encoder reranking, before and after independent review.	54
5.8	Historical baselines re-scored on the frozen benchmark. Few-shot and RAG were regenerated against the frozen question wording and cover all 98 records, with citation measures over the 88 whose gold citation resolves to a provision. LoRA retains its stored predictions and is therefore scored on the 73 records whose wording did not change, 63 of them resolvable. The loose measure is the original scorer, reproduced verbatim and reported for continuity only.	57
5.9	Where the change from the earlier draft's figures comes from, at parent level. <i>Stored</i> , 73 is the draft's comparison. <i>Regenerated</i> , 73 holds the record set fixed and changes only the predictions, so it measures run-to-run variation. <i>All 98</i> holds the provenance of the predictions fixed and adds the 25 records the draft excluded. Denominators are 63 and 88 resolvable records.	57
5.10	Effect of a single prompt rule forbidding elided quotes, and how it transfers to unseen questions.	61
5.11	Compact against enriched context on the development partition ($n = 42$). Prompt, schema, model, temperature and question set are identical; only the context construction differs.	62
5.12	Candidate escalation rules on the development partition ($n = 42$, 9 error events). Precision intervals are Wilson 95%.	64
5.13	Sample size required to demonstrate a given escalation precision against the observed 21.4% base error rate.	64
5.14	Resolution of the benchmark's own gold citation labels against the corpus ($n = 98$).	67
A.1	Environments used, as recorded in the experiment manifests.	84
A.2	Hardware and principal library versions for environment B, the machine on which most of the work was performed.	85
A.3	Completion requests and tokens, from the provider's usage export. Consolidated in <code>docs/billing/</code>	86
A.4	Embedding tokens, from the stored cache metadata, deduplicated on the text digest.	86
A.5	Recorded charges, from the provider's cost export.	87

B.1	AI-assisted artefacts, from the artefact log. “Verified” states what the author checked before the artefact was accepted.	90
-----	---	----

List of Abbreviations and Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Matching 25
DGAC	Direction générale de l’Aviation civile
DSAC	Direction de la sécurité de l’Aviation civile
EASA	European Union Aviation Safety Agency
ERAS	Évaluation des risques et analyses de sécurité
EU	European Union
GM	Guidance Material
GPU	Graphics Processing Unit
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MEAS	Mission évaluation et amélioration de la sécurité
MRR	Mean Reciprocal Rank
nDCG	normalised Discounted Cumulative Gain
NLP	Natural Language Processing
PFE	Projet de Fin d’Études
PGE	Programme Grande École
QA	Question Answering
RAG	Retrieval-Augmented Generation
RRF	Reciprocal-Rank Fusion

Chapter 1

Introduction

1.1 Background and Motivation

European aviation safety rests on a layered body of binding regulation, implementing acts, annexes, and guidance material. A person answering a regulatory question must identify not merely a plausible rule but the applicable instrument, the article, the paragraph, the version in force, and the conditions and exceptions that surround it. Two provisions may state compatible obligations while only one of them governs a given operator; a requirement may be qualified by an exception three subparagraphs later; and guidance material may explain how a requirement is commonly met without itself imposing anything.

This makes aviation-regulation question answering fundamentally different from open-domain conversational assistance. Fluency is not merely insufficient — it is actively dangerous, because a fluent answer that cites the wrong provision is harder for a non-specialist to detect than an obviously uncertain one. A useful answer must be supported by authoritative text and traceable to a precise citation that a qualified reader can check in a few seconds.

Large Language Models (LLMs) create a genuine opportunity here. Regulatory material is voluminous, cross-referenced, and difficult to navigate for anyone who is not a full-time specialist, and a system that answers in natural language while pointing at the governing provision would meaningfully reduce the cost of finding an answer. But the same technology introduces a reliability problem that the aviation domain cannot absorb. A model can produce a confident explanation when it has retrieved the wrong provision, or when the cited source does not entail the claim built on it.

Retrieval-Augmented Generation (RAG) reduces this risk by placing retrieved evidence in the generation context. It does not eliminate it. A recent audit of commercial retrieval-augmented legal research tools found that they still produce unsupported or miscited output in a substantial fraction of responses [1], and a human evaluation of generative search engines found that only about half of generated sentences were fully supported by their own citations, with fluency correlating *inversely* with citation precision [2]. Grounding is an improvement, not a safety property.

There is a second and less-discussed difficulty. “RAG” names a family of separable decisions rather than a single intervention: source parsing, chunk boundaries, retrieval model, candidate depth, ranking, context construction, prompt contract, citation handling, and verification policy [3]. Each of these can fail independently, and each failure mode calls for a different response. A single end-to-end accuracy figure conceals which component failed, and therefore conceals what should be done about it.

1.2 Problem Statement

The initial phase of this project compared three approaches over the same regulatory material: few-shot prompting, a retrieval-grounded pipeline, and a model adapted with Low-Rank Adaptation (LoRA). The retrieval-grounded system performed best. That comparison remains an informative baseline, but two problems prevent it from supporting the conclusions it appears to support.

The first is confounding. The corpus of that phase represented the source material as comparatively flat chunks, some longer than 30,000 characters, while the generator received only the first 1,500 characters of the selected chunk. A retrieval evaluation could therefore count a source as found when the supporting paragraph never reached the generator at all. The benchmark itself consisted of synthetic questions whose reference answers had not been checked against the provisions they cited, citation matching was performed by substring comparison so that a partial match to the wrong subparagraph scored half credit, and document identity was not preserved across the several instruments consolidated in the source publication. Treating every observed error as an intrinsic limitation of retrieval-augmented generation would confound system failure with data and evaluation failure.

The second is that a leaderboard does not answer the operational question. Knowing that one architecture scores higher than another says nothing about what a deployed system should do when it is about to answer badly. In a compliance setting that second question is the one that matters, because the acceptable behaviour on an unanswerable question is not a low-confidence guess — it is a refusal and a referral.

This work therefore repairs the confounders first, and then reframes the contribution around failure diagnosis and calibrated escalation.

Primary research question. Where and why does retrieval-augmented generation fail in citation-bearing aviation-regulation question answering, and which observable conditions justify escalating from standard RAG to more advanced retrieval, verification, multi-step reasoning, or human review?

The primary question is decomposed into six operational questions:

1. How much error originates in corpus parsing, retrieval, ranking, context construction, generation, or citation verification?
2. Does a hierarchy-aware parent–child representation improve evidence retrieval and exact citation traceability?
3. Does a larger dense embedding model provide a meaningful gain over the smaller baseline under an identical corpus and query set?
4. Does hybrid lexical and dense retrieval recover terminology-heavy and semantically paraphrased questions more reliably than either channel alone?
5. When does cross-encoder reranking or local context expansion recover evidence already present in a broader candidate set?
6. Under which conditions should the system clarify, abstain, or refer the question to a qualified human rather than apply another automated step?

The sixth question is the one that carries the contribution. The first five are instrumental to it: each identifies a class of failure, and the escalation policy is the answer to what should be done when that class of failure is detected at runtime.

1.3 Methodological Stance

Three commitments shape how the work is conducted, and they are stated here because they explain several choices that would otherwise look conservative.

The measuring instrument is validated before its measurements are believed. A benchmark whose gold labels are partly machine-generated cannot certify itself. Before any performance figure in this thesis is described as final, a stratified sample of the evaluation set is audited by human reviewers and the agreement between them is reported rather than asserted. This is why the retrieval results in Chapter 5 carry a candidate status: not because they are unreliable, but because the standard of evidence for a final claim has not yet been met.

One component changes at a time. Where two changes are of interest, they are run as separate experiments with separate manifests. This costs experimental throughput and buys attributability, which is the only thing that makes a failure analysis possible.

A negative result is retained. Configurations that were tried and rejected — unequal fusion weights, parent-mass score boosting — remain documented rather than quietly removed, so that the reported design can be read against the alternatives that were considered.

1.4 Objectives and Contributions

The project pursues five measurable objectives. First, it establishes a reproducible evidence corpus in which every retrievable unit is linked to a coherent legal parent, a source document, a hierarchy, and a canonical citation. Second, it audits the legacy benchmark rather than silently treating synthetic source labels as ground truth. Third, it compares retrieval interventions through controlled ablations that hold the corpus, questions, and split policy fixed. Fourth, it separates evidence retrieval from answer generation and from citation verification, so that each can fail visibly and independently. Fifth, it converts the resulting failure taxonomy into a switching policy driven by observable runtime signals rather than by gold labels that are unavailable in deployment.

The principal technical and methodological contributions are:

- a deterministic parent–child representation of the stored aviation-regulation sources, preserving instrument identity and the distinction between binding text and guidance material;
- exact lineage from historical question identifiers to parent and child evidence units, with fuzzy matches never silently accepted;
- parent-grouped development and test partitions that prevent evidence leakage between tuning and reporting;
- a controlled comparison of two dense embedding models, local BM25, reciprocal-rank fusion, and a local cross-encoder reranker, with model selection performed on development data only;
- structured generation using application-resolved citations and verbatim evidence quotes, so that a citation cannot be fabricated by the model;
- a pre-registered multi-reviewer validation protocol for the benchmark, with chance-corrected agreement reported and contested records listed individually;
- a distinction, made empirically rather than assumed, between ranking failures, candidate-generation failures, ambiguous questions, incomplete benchmark labels, unsupported generation, and evidence conflict;
- a cost-aware escalation ladder that reserves multi-step or agentic retrieval for residual cases that have been validated as genuinely requiring it.

The seventh of these deserves emphasis, because it is the finding that reframed the project. Three questions in the evaluation set resisted retrieval even at a candidate depth of twenty, and it would have been natural to report that as a system ceiling.

Examined individually they were three different problems, and the independent review later settled which was which: one question was defective and no reviewer accepted it, while the panel accepted the other two, making those genuine retrieval failures rather than label defects. An earlier internal reading had attributed one of them to incomplete gold evidence; the reviewers disagreed, and their verdict stands. Reporting all three as a single ceiling would have been a measurement error rather than a result, and that observation is what motivated the validity protocol that now governs the whole evaluation.

1.5 Scope and Delimitations

The empirical source material is restricted to the aviation-regulation documents stored in the project workspace: Regulation (EU) No 376/2014 on occurrence reporting, together with its implementing and amending instruments and the associated guidance material, as consolidated in the European Union Aviation Safety Agency (EASA) Easy Access Rules publication [4], [5]. No claim is made about generalisation to other regulatory instruments; the corpus, benchmark, and protocol are designed to be transferable, but transfer is not demonstrated here.

The present benchmark evaluates English regulatory evidence. French-to-English retrieval, bilingual answer generation, and a bilingual corpus are realistic extensions and are treated in the state of the art, but they are not mixed into the current evaluation without an explicit protocol, because doing so would confound a language effect with the retrieval effects under study.

Three further delimitations apply. The three-system baseline is an end-to-end comparison, not a causal claim about architectures: the systems do not share a base model, so any difference between them confounds architecture with model capability. The generation stage evaluates a commercial model that cannot be fine-tuned, so all adaptation in this work is on the retrieval side. And the evaluation set, while grounded in real regulatory text, consists of synthetic questions audited by human reviewers rather than questions independently authored by domain practitioners.

Finally, the study evaluates a research prototype, not a certified legal decision system. Generated answers cannot replace competent legal or aviation-safety review. Where evidence is absent, ambiguous, or conflicting, abstention and human referral are intended outcomes rather than system failures — a position that is not a disclaimer but a design commitment, and one that is tested as a safety property in Chapter 4.

1.6 Report Structure

Chapter 2 reviews the research this work builds on and the research that defines what an adequate answer to the problem would look like: retrieval-augmented generation, lexical and dense retrieval, granularity and reranking, legal and regulatory language processing, attribution and verifiability, selective answering, and the evaluation methodology through which any of it can be measured.

Chapter 3 presents the professional context, the mission and its revision, the regulatory domain, the data and its provenance, and the constraints under which the project operates.

Chapter 4 describes the corpus redesign, the benchmark audit and its validity gate, the retrieval architecture, the generation contract, verification and selective answering, the escalation policy and its calibration, the evaluation measures, and the reproducibility controls.

Chapter 5 reports the frozen benchmark and the review that produced it, retrieval and reranking on reviewed labels, the historical baselines re-scored under a corrected citation evaluator, evidence-constrained generation with its verification, and the calibration of the escalation policy.

Chapter 6 answers the research question to the extent that the reviewed evidence supports, and states the limitations and the work that remains.

Chapter 2

State of the Art

2.1 Scope and Organisation

This chapter reviews the research that the present work builds upon and, more importantly, the research that defines what would count as an adequate answer to the problem it addresses. The review is organised around the pipeline under study rather than around chronology. Section 2.2 establishes why an external evidence store is required at all, and Section 2.3 examines the competing option of adapting model weights instead. Sections 2.4 to 2.6 treat the retrieval stack: lexical, dense, and hybrid candidate generation, the choice of indexing unit, and cross-encoder reranking. Section 2.7 turns to the specific properties of legal and regulatory language and to the empirical record of automated systems operating on it. Sections 2.8 and 2.9 cover attribution and selective answering, which together constitute the safety layer of the architecture. Sections 2.10 and 2.11 address measurement: what to measure, and how to know that the measuring instrument is itself sound. Section 2.12 treats multi-step and agentic retrieval, and Section 2.13 states the gap that this work occupies.

Two commitments shape the selection of material. First, a claim about system performance is only admitted here if the cited work reports how it was measured; several widely repeated assertions about retrieval-augmented generation do not survive that filter. Second, work that documents *failure* is treated as at least as important as work that reports improvement, because the contribution of this project is a policy for responding to failure rather than a new architecture.

2.2 From Parametric Knowledge to Retrieval-Augmented Generation

Transformer architectures made it possible to model long-range dependencies through attention rather than recurrence [6], and bidirectional pre-training subsequently improved language understanding and transfer across many Natural Language Processing (NLP) tasks [7]. Knowledge acquired this way is stored implicitly in model weights. That storage mechanism has three properties that are unacceptable in a regulatory setting:

it is not addressable, so a statement cannot be traced to the text that licensed it; it is not updatable, so an amended provision cannot be corrected without retraining; and it is not auditable, so a reader cannot verify a claim without independently rediscovering its source.

RAG addresses these properties by conditioning generation on passages retrieved from an external collection at inference time. Lewis et al. jointly modelled retrieval and sequence generation for knowledge-intensive tasks [8], and REALM integrated retrieval directly into language-model pre-training so that the retriever was learned rather than fixed [9]. Subsequent architectures explored how retrieved evidence should enter the generator. Fusion-in-Decoder encodes each passage independently and allows the decoder to attend jointly across all of them, with accuracy improving as more passages are supplied [10]. RETRO takes the scaling argument further, showing that a 7.5-billion-parameter model with chunked cross-attention over a two-trillion-token retrieval database can match models an order of magnitude larger [11], which establishes that capability can be relocated from weights into an index. REPLUG inverts the usual adaptation target by treating the language model as a frozen black box and training the retriever against its feedback [12]; this matters here because the generator in the present system is a commercial Application Programming Interface (API) that cannot be fine-tuned, so all available adaptation is on the retrieval side.

The literature has since consolidated into a recognised taxonomy of naive, advanced, and modular RAG, decomposing the pipeline into indexing, retrieval, augmentation, and generation stages each with characteristic failure modes [3]. That decomposition is adopted throughout this work. It is worth stating plainly that “RAG” is not a system but a family of separable decisions: corpus construction, chunking unit, retrieval model, candidate depth, reranking, context assembly, prompt contract, and verification policy. Reporting a single accuracy figure for “RAG” without specifying these choices communicates very little, and comparisons across papers that differ on several of them simultaneously are not informative about any one of them.

Adaptive retrieval marks the current frontier. Self-RAG trains a model to emit reflection tokens deciding when retrieval is necessary and then to critique whether its own output is supported by the retrieved evidence, improving both factuality and citation precision over fixed-retrieval baselines [13]. This is the closest published antecedent to the escalation policy developed in Chapter 4, and the difference is instructive: Self-RAG learns the control decision inside the model, whereas the present work externalises it into an inspectable policy over observable retrieval signals, precisely so that the decision can be audited by a compliance reviewer.

In a high-stakes regulatory setting the objective is stricter than general factual usefulness. The selected evidence must identify the governing legal unit; each material claim must be entailed by that evidence; and the system must recognise when a question

is ambiguous or when no retrieved source supports a safe answer.

2.3 Retrieval or Parametric Adaptation?

The alternative to retrieval is to write domain knowledge into the weights. Parameter-efficient methods have made this affordable. Low-Rank Adaptation freezes the pre-trained weights and trains only injected low-rank matrices, reducing trainable parameters by several orders of magnitude with no additional inference latency [14], and quantised adaptation backpropagates through a frozen four-bit model into those adapters, bringing the fine-tuning of very large models within reach of a single consumer Graphics Processing Unit (GPU) [15]. The baseline experiment reported in Chapter 5 uses exactly this route, and it is therefore not open to this work to dismiss fine-tuning on cost grounds.

The comparison must instead be made on knowledge properties. Ovadia et al. compare unsupervised fine-tuning against retrieval for injecting new and time-shifted knowledge across knowledge-intensive benchmarks and find retrieval consistently superior, with fine-tuning struggling in particular on facts absent from pre-training [16]. For a regulatory corpus the argument is sharper still. Regulation (EU) No 376/2014 has been amended since adoption [4], and the consolidated presentation used here is a dated revision [5]; an adapter trained on one revision silently encodes it. More decisively, adapter weights cannot supply provenance. A fine-tuned model may produce a correct obligation and remain unable to identify which paragraph imposes it, which is the property the application actually requires.

The correct reading of the three-system baseline in this thesis is therefore not that fine-tuning is weak, but that the comparison is between architectures with different *knowledge-access* properties, only one of which supports citation. This framing is stated here because the baseline results are otherwise open to a misreading that the earlier project documentation did not adequately guard against.

2.4 Sparse, Dense, and Hybrid Retrieval

Sparse retrieval represents queries and documents through lexical statistics. BM25 remains a strong baseline because it rewards exact term overlap while normalising for document length [17]. This is valuable in regulations, where article numbers, acronyms, defined terms, and formulaic expressions carry substantial meaning: a query naming Article 4(6) is not well served by a model that treats the numeral as noise. Learned sparse retrieval shows that the lexical channel is itself improvable; SPLADE learns sparse, expansion-aware term weights in the vocabulary space of a pre-trained encoder, retaining inverted-index efficiency while mitigating vocabulary mismatch [18]. The vocabulary-mismatch weakness that motivates a dense channel is therefore a property

of BM25 specifically rather than of sparse retrieval as such, and this qualification is carried through the experimental discussion.

Dense retrieval maps queries and passages into a learned vector space. Dense Passage Retrieval demonstrated that independently encoded questions and passages support efficient open-domain retrieval [19], and Sentence-BERT showed how siamese encoders make large-scale semantic similarity search computationally practical [20]. Contriever established that contrastive pre-training on unlabelled text alone yields a dense retriever competitive with BM25 across the BEIR suite, and that it transfers across languages [21]; this supports the use of a general-purpose commercial embedding model as a zero-shot first-stage retriever without in-domain training. Between these single-vector models and full cross-attention lies late interaction: ColBERT encodes query and document tokens separately and defers their comparison to a cheap maximum-similarity operator [22], and ColBERTv2 adds distillation and residual compression to cut the index footprint substantially while improving quality [23]. Late interaction is not used in this work; it is recorded here as a deliberate omission, traded away for a simpler index and no bespoke training.

Multilingual embedding models bear directly on the scope question raised in Chapter 3. M3-Embedding produces dense, lexical, and multi-vector representations across more than one hundred languages from a single model, and reports that combining its dense and sparse scoring modes outperforms either alone [24]. That internal result is independent corroboration of the premise underlying hybrid retrieval. Model selection across the embedding market is now conventionally argued through the Massive Text Embedding Benchmark, which showed that no single embedding model dominates across task families [25] — a finding that justifies selecting an embedding model against the target task rather than against a general leaderboard position.

Hybrid retrieval treats lexical and semantic evidence as complementary. Reciprocal-rank fusion combines ranked lists without requiring their raw scores to be calibrated [26]. For a document d retrieved by systems i , the fused score is

$$\text{RRF}(d) = \sum_i \frac{1}{c + \text{rank}_i(d)}, \quad (2.1)$$

where c is a fixed rank constant. The method is attractive for controlled experiments because it has a small and interpretable parameter surface and requires no score normalisation.

That attractiveness must be qualified. Bruch et al. analyse fusion functions for hybrid retrieval theoretically and empirically, and report that a convex combination of normalised scores outperforms Reciprocal-Rank Fusion (RRF) both in and out of domain, noting that a rank-based method necessarily discards score magnitude and is sensitive to its rank constant [27]. The present work nonetheless adopts equal-

weight RRF with a fixed constant, and the justification is methodological rather than performance-based: a fusion function with no tuned parameters cannot leak information from the held-out split, which matters more here than the last increment of recall. This trade-off is stated explicitly in Chapter 4 and revisited among the limitations.

Table 2.1 sets the families side by side on the property that matters for regulatory text.

Table 2.1 – Retrieval families and their principal trade-offs, as characterised by the cited work.

Family	Representative	Property relevant to regulatory text
Lexical sparse	BM25 [17]	Rewards exact identifiers and formulaic wording; blind to paraphrase.
Learned sparse	SPLADE [18]	Adds term expansion while remaining servable on an inverted index.
Single-vector dense	DPR [19], Contriever [21]	Recovers paraphrase; may underweight exact article references.
Late interaction	ColBERT [22], [23]	Higher accuracy at greater index cost; not adopted here.
Multilingual multi-mode	M3-Embedding [24]	Single model spanning languages and scoring modes.
Rank fusion	RRF [26]	No score calibration, no tuned parameters; discards magnitude [27].

2.5 Evidence Granularity and Document Hierarchy

The choice of indexing unit is an independent lever on retrieval quality rather than an implementation detail. Chen et al. compare passage-, sentence-, and proposition-level indexing and find the finest granularity consistently strongest, with the largest gains on entities that are rare in the training distribution [28]. Regulatory text amplifies this effect, because the normative content of a provision is typically carried by a single subparagraph while the surrounding article supplies scope and definitions. A coarse chunk containing several obligations dilutes the signal for each of them.

Fine granularity introduces the opposite hazard. A subparagraph detached from its heading, its defining article, and its table header may be uninterpretable, and in a legal document it may be actively misleading — an exception read without its governing rule inverts the obligation. A parent–child design addresses this tension by retrieving atomic children while retaining links to coherent legal parents, so that context can be restored after ranking rather than before it.

Context restoration must nevertheless be bounded, and the reason is empirical. Liu et al. document a U-shaped performance curve in long contexts: models use evidence

reliably at the beginning and end of the supplied window but degrade sharply when the relevant passage sits in the middle, with performance falling as the context grows even for models advertised as long-context [29]. Two consequences follow for this work. First, the alternative design of supplying the entire regulation to a long-context model is not a safe default. Second, ordering and context length are substantive variables, which is what makes reranking worth its cost and what motivates a tight evidence budget rather than a generous one.

This project therefore treats hierarchy as explicit metadata rather than as a licence to promote every child of a long parent. Parent-aware expansion is applied only after retrieval and within a bounded context budget. The distinction matters because hierarchy can improve interpretability without necessarily improving relevance ranking, and the two effects must be measured separately.

2.6 Cross-Encoder Reranking

Bi-encoder retrieval is efficient because document vectors can be pre-computed and cached, but query and passage interact only through a similarity function applied to representations formed in ignorance of one another. A cross-encoder instead processes the pair jointly, allowing full attention between query and candidate terms. BERT-based passage reranking demonstrated substantial gains from this direct interaction [30], and monoT5 showed that the effect is not an artefact of encoder-only models by recasting ranking as sequence-to-sequence target-token generation, with the additional finding that sequence-to-sequence rerankers degrade more gracefully in low-data regimes [31]. Reranking is therefore an architectural stage rather than a property of one model family.

The cost of full attention is tolerable only over a small candidate pool, which is what makes distillation relevant. MiniLM compresses a large transformer by matching the teacher’s self-attention distributions and value relations, producing students that retain the large majority of teacher performance at a fraction of the cost [32]. The reranker used in this work is a MiniLM cross-encoder trained on MS MARCO, a large collection of anonymised web search queries with human-generated answers and passage judgements [33]. Naming that training distribution is not a formality. MS MARCO queries are short, factoid-oriented, and drawn from general web search; questions about regulatory obligations are longer, structurally explicit, and drawn from a specialist sublanguage. Any gain observed from this reranker is therefore obtained *despite* a domain shift, and any residual failure must be interpreted with that shift in view.

Reranking and candidate generation solve different problems, and conflating them is a recurrent source of confusion in applied reports. A reranker can move a relevant passage from rank 18 to rank 3; it cannot recover a passage absent from the candidate pool. Candidate recall at the fusion depth is thus a hard ceiling on the reranked result,

and the two quantities must be reported separately. This distinction is central to the switching policy developed in this work.

2.7 Legal and Regulatory Language Processing

Legal text is a genuine distribution shift rather than ordinary domain vocabulary. LEGAL-BERT showed that domain-adaptive pre-training on legislation and case law yields consistent downstream gains, and that the standard fine-tuning recipe does not transfer unmodified to specialist domains [34]. CaseHOLD refined the condition under which such adaptation pays: gains materialise when the task is both sufficiently domain-specific and sufficiently difficult, and are marginal otherwise [35]. LexGLUE subsequently standardised evaluation across seven English legal datasets, including tasks defined over EU legislation [36].

These benchmarks are informative about classification and, read carefully, they are also informative about what is missing. They score aggregate accuracy over labels; none of them asks whether a generated statement is attributable to a specific provision. LegalBench makes the gap visible from a different direction. Built collaboratively by legal professionals around a typology of legal reasoning, it shows that model performance is highly uneven across reasoning types, with issue-spotting comparatively strong and rule application and conclusion comparatively weak [37]. A single accuracy figure therefore conceals a task-dependent competence profile, which is precisely the information a compliance deployment requires in order to decide where a human must remain in the loop. SARA makes the same point on statutory material specifically: pairing tax statutes with entailment and numerical cases, it shows that models trained on general corpora apply rule text to fact patterns poorly [38]. Questions over aviation regulation are rule-application questions of exactly this kind, so retrieval quality cannot be assumed to entail correct application.

The most directly relevant empirical result concerns deployed systems. Magesh et al. audit commercial retrieval-augmented legal research tools and find that grounding reduces but does not eliminate hallucination, with error rates in the range of roughly one in six to one in three responses, including miscited and unsupported authority [1]. Their distinction between an answer that is wrong and an answer that is right but misgrounded is adopted directly in this thesis. It is the empirical basis for the claim that retrieval grounding is not by itself a safety property, and that a citation-bearing system must be evaluated on groundedness rather than on a headline accuracy figure.

Aviation regulatory text has attracted its own resources. *aeroBERT* constructed an annotated corpus of aerospace requirements drawn from certification regulations and showed that a domain-adapted encoder substantially outperforms general-purpose zero-shot classifiers on it [39], establishing the precedent for treating aviation regulation as a

distinct NLP domain with its own resources. Related work on occurrence narratives shows that free-text safety reports carry signal absent from structured fields [40]. That literature addresses the reports produced *under* an occurrence-reporting regime; the present work addresses the regime itself, that is, the regulation that determines what must be reported, by whom, and within what deadline [4], [5]. To the author’s knowledge no published benchmark targets citation-bearing question answering over this instrument, which is the concrete gap the corpus and benchmark of Chapter 4 are built to fill.

2.8 Attribution, Citation, and Verifiability

Hallucination is standardly divided into intrinsic cases, where output contradicts the source, and extrinsic cases, where output is simply unverifiable against it [41]; more recent surveys reframe the same territory as a distinction between factuality and faithfulness and trace causes through data, training, and inference [42]. In a regulatory application the extrinsic case dominates the risk profile, because a fluent statement that no provision supports is both plausible to a non-specialist reader and undetectable without checking the cited text.

Attribution research supplies the formal apparatus for that check. The Attributable to Identified Sources framework defines a generated statement as attributable only if a generic hearer would affirm “according to source X , S ” given the cited evidence, and supplies an annotation protocol demonstrating that the property is measurable rather than intuitive [43]. Attributed question answering operationalises the criterion into a task of answer-plus-supporting-passage, and its architectural comparison finds retrieve-then-read pipelines superior to post-hoc citation attachment on both correctness and attribution [44]. That result is the justification for constraining generation to retrieved source identifiers at generation time rather than adding citations afterwards.

The scale of the problem in deployed systems is documented directly. Liu et al. human-audit four commercial generative search engines and find that only about half of generated sentences are fully supported by their own citations, and that roughly a quarter of citations do not support the statement they are attached to, while fluency and perceived utility correlate *inversely* with citation precision [2]. Citation presence and citation support are thus distinct quantities, and only the former is cheap to measure. ALCE turned this observation into a reproducible benchmark, scoring fluency, correctness, and citation precision and recall separately over retrieval corpora, and confirming that strong models leave a substantial fraction of statements uncited or wrongly cited [45]. RARR offers the complementary repair path: rather than regenerating, retrieve evidence per claim and minimally revise unsupported spans, trading attribution against preservation of the original output [46]. This is the published

antecedent for treating a verification failure as a trigger for targeted revision rather than as an immediate abstention.

The architecture examined in this work applies structural safeguards derived from this literature: the generator returns source identifiers rather than free-form citation strings, supporting quotes must occur verbatim in the supplied evidence, and the application resolves identifiers to canonical citations. Structural checks are deterministic and cheap, but they do not establish semantic entailment. A verbatim quote may be accurate and still fail to support the claim built on it, and material omission is invisible to a quote check. Claim support must therefore be assessed separately, and the evaluation in Chapter 5 reports the two families of check independently.

2.9 Selective Answering, Calibration, and Escalation

A system that must answer every question cannot be safe on a corpus where some questions are ambiguous and some answers are unsupported. Selective prediction provides the formal treatment: a predictor is paired with a rejection function, and performance is characterised by a risk–coverage curve rather than by a single accuracy figure, with distribution-free guarantees available at a chosen coverage [47]. Framing abstention this way converts an apparently ad hoc threshold into a defensible operating point.

The signal driving that decision requires care. Kamath et al. show that model confidence is badly miscalibrated under domain shift and that a separate calibrator trained on mixed in-domain and out-of-domain data substantially outperforms thresholding the model’s own softmax [48], which argues for separating the answering model from the abstention decision. Earlier work on question-answering calibration reached a compatible conclusion, finding models poorly calibrated out of the box and improvable by fine-tuning, temperature scaling, and candidate paraphrasing [49]. Against this, Kadavath et al. show that large models are reasonably well calibrated in suitable formats, can self-evaluate sampled answers, and can be trained to predict whether they know an answer, with partial generalisation to unseen tasks [50]. Taken together, model-internal confidence is usable as *one* input to an escalation trigger and unsafe as the only one — which is why the policy developed here combines retrieval disagreement, score margin, and verifier outcome rather than relying on a generator score.

Abstention can also be learned directly. R-Tuning diagnoses standard instruction tuning as forcing an answer regardless of parametric knowledge, then partitions training data by what the model demonstrably knows and fine-tunes explicit refusal on the uncertain partition, improving both accuracy and abstention and showing that refusal

behaves as a transferable skill [51]. This work does not train refusal, because the generator is not trainable in the deployed configuration; the result is recorded because it establishes that a rule-based abstention gate is a design choice with a learned alternative, not the only available mechanism.

2.10 Evaluating Retrieval and Generation

Retrieval evaluation is comparatively settled. Recall at a fixed depth, Mean Reciprocal Rank (MRR), and normalised Discounted Cumulative Gain (nDCG) are standard, the last introduced to credit graded relevance and to discount gain by rank position [52]. BEIR established that retrieval models behave very differently across tasks and domains, so a model selected on one benchmark cannot be assumed to transfer [53]. KILT is the closer methodological precedent for this work: it unifies knowledge-intensive tasks over a single corpus snapshot and mandates provenance annotations, so that systems are scored on retrieved evidence as well as on final output [54].

Generation evaluation is less settled and more consequential. ROUGE measures recall-oriented n-gram overlap against reference text [55] and remains widely reported, but Maynez et al. demonstrate through large-scale human annotation that such overlap metrics correlate only weakly with human faithfulness judgements, while entailment-based measures correlate considerably better [56]. A ROUGE score therefore cannot license a conclusion about factual correctness. This is not a marginal caveat in the present setting: the earlier phase of this project reported ROUGE-L alongside a citation-matching proxy, and the reinterpretation of those figures in Chapter 5 rests directly on Maynez et al.

Purpose-built RAG evaluation frameworks decompose quality along the pipeline. RAGAS scores faithfulness, answer relevance, and context relevance without reference answers, using prompted model judgements [57]. ARES trains lightweight judges on synthetic data and then corrects their bias using prediction-powered inference over a small human-annotated set [58]. ARES is the most directly applicable precedent for the design adopted here, because it addresses the same structural problem: a largely machine-generated evaluation resource made trustworthy by a human-anchored correction, with the uncertainty of that correction reported rather than suppressed.

Both frameworks depend on model-generated judgements, which imports a known set of biases. Zheng et al. characterise language models as judges systematically, documenting position, verbosity, and self-enhancement bias alongside weak grading of reasoning, while also showing agreement with human judgements at a level comparable to human-human agreement [59]. The correct inference is neither to reject automated judging nor to accept it silently, but to report the human-agreement figure that validates it in the specific setting of use.

2.11 Benchmark Validity and Reviewer Agreement

An evaluation is only as trustworthy as its gold data, and this is where the present work makes an unusually large methodological investment. Two threats are relevant.

The first is contamination. Sainz et al. argue that closed pre-training corpora have absorbed the standard public benchmarks, that this invalidates measured progress, and that contamination must be documented per benchmark rather than assumed absent [60]. Building a fresh benchmark over a specific regulatory instrument mitigates this threat by construction, at the cost of introducing the second one.

The second is the validity of machine-generated evaluation items. Where questions and reference answers are produced by a model from the same family that is later evaluated, agreement between system and gold may reflect shared provenance rather than correctness. The mitigation adopted in this work is human audit of a stratified sample, with agreement reported rather than asserted. That requires chance-corrected agreement statistics: Cohen’s coefficient for two raters on nominal categories [61], Fleiss’s generalisation for a fixed number of raters drawn from a larger pool [62], and the conventional verbal benchmarks for interpreting the resulting magnitudes [63]. Landis and Koch present those cut-offs as arbitrary conventions, and they are treated as such here.

Reported proportions require intervals. At the sample sizes typical of a human-audited subset, and at proportions near zero or one, the normal-approximation interval is inappropriate; the score-based interval obtained by inverting the normal test on the true proportion remains within the unit interval and is used throughout [64].

A final observation belongs here rather than in the results. Where independent qualified reviewers disagree about whether an answer is supported by a provision, that disagreement is evidence about the question, not noise in the measurement. The literature on legal reasoning benchmarks acknowledges genuine indeterminacy in rule application [37], [38], and an evaluation protocol that resolves every disagreement by majority vote discards exactly the signal that identifies ambiguous questions. This work therefore reports contested records separately.

2.12 Agentic and Multi-Step Retrieval

Multi-step retrieval can decompose a question, follow cross-references, or issue a revised query after inspecting initial evidence. Self-RAG demonstrates the value of learning when to retrieve at all [13], and post-hoc revision architectures show that a failed verification need not force abstention [46]. These capabilities are genuinely useful when a validated residual requires several sources, which is a realistic prospect in regulation, where obligations are frequently defined by cross-reference.

They also introduce latency, monetary cost, and additional failure modes, and they complicate attribution: an answer assembled over several retrieval rounds is harder to trace to a single governing provision. Applying multi-step retrieval to every question would obscure whether a simpler retriever, a reranker, a clarification request, or an abstention rule was sufficient. The position taken in this work, and recorded in the decision log, is that agentic retrieval is admitted only after a residual failure has been shown to require multi-step evidence following independent review of the benchmark. That ordering — diagnose first, escalate second — is the same principle the contribution advances at the level of individual questions.

2.13 Positioning of This Work

The literature reviewed above supports each component of the pipeline individually. Hybrid retrieval is well motivated [24], [26], [27]; reranking is well established [30], [31]; granularity effects are measured [28], [29]; attribution is formalised and its failure in deployed systems is documented [1], [2], [43]; selective answering has a statistical foundation [47], [48]; and evaluation frameworks exist for each stage [54], [57], [58].

Three gaps remain, and together they define this work.

First, these components are usually evaluated in isolation and on general-domain corpora. There is no published study that assembles them over a single European aviation-regulation instrument and reports, for the same questions, where each stage succeeds and fails.

Second, the interventions are typically presented as improvements to be applied uniformly. The question of *when* an intervention is warranted — which observable runtime condition justifies reranking, expansion, clarification, or human referral for a particular question — is largely unaddressed. Self-RAG learns this control internally [13]; a compliance setting needs it externalised and inspectable.

Third, and most consequentially, the field reports system failure against gold data whose own validity is rarely established. Where an apparent retrieval ceiling is in fact a mixture of genuine system failure, incomplete gold evidence, and irreducibly ambiguous questions, reporting it as a ceiling is a measurement error rather than a finding.

Table 2.2 maps each level of the escalation ladder used in this work to the prior work that motivates it.

The contribution of this thesis is accordingly not a new foundation model, nor a new retrieval architecture. It is a controlled, evidence-first study of where a regulatory RAG pipeline fails and how a system should respond, conducted under a protocol that establishes the validity of its own gold data before quoting any ceiling. The work combines hierarchy-aware corpus construction, hybrid retrieval, local reranking, explicit evidence contracts, audited benchmark construction with reported reviewer

Table 2.2 – Escalation levels and the prior work that motivates each. The contribution of this thesis is the calibrated transition between levels, not the levels themselves.

Level	Intervention	Motivating prior work
0	Dense retrieval over flat chunks	[8], [19]
1	Hierarchy-aware parent–child retrieval	[28], [29]
2	Hybrid lexical and dense candidates with fusion	[17], [26], [27]
3	Cross-encoder reranking of the candidate pool	[30]–[32]
4	Evidence-constrained generation and verification	[43]–[45]
5	Clarification or targeted revision	[46]
6	Abstention and referral to human review	[1], [47], [48]

agreement, and selective escalation in one reproducible protocol. Its evaluation separates data defects from retrieval, ranking, generation, and verification failures, so that each intervention is justified by a diagnosed failure mode rather than adopted because it is available.

Chapter 3

Project Context and Scope

3.1 Host Organisation and Team

This project was carried out during an apprenticeship at the French **Direction générale de l’Aviation civile** (DGAC), within the **Direction de la sécurité de l’Aviation civile** (DSAC), in the **Mission évaluation et amélioration de la sécurité** (MEAS) and specifically its **Évaluation des risques et analyses de sécurité** (ERAS) division.

The placement matters to this work for one reason: ERAS is, in the French context, an office that operates the regime this system answers questions about. Its remit covers the identification of hazards to civil aviation, the collection and analysis of occurrence reports, contribution to the State Safety Programme, and liaison with the safety planning of the EASA. Regulation (EU) No 376/2014 is not an abstract object of study there; it is the instrument under which the work of the division is conducted. The practitioners whose reading of citation labels is reported in Chapter 5 are people who work with this regime professionally, and the observation that some of those labels could not be resolved carries more weight for having come from them.

What this project is, and what it is not. Two statements are necessary for the reader to place this work correctly.

First, **the assigned apprenticeship mission was distinct from the subject of this mémoire.** The day-to-day assignment was software development in Python on other matters. This project is a Final Year Project undertaken in the context of that placement and informed by it; it was not a deliverable requested by the division, and nothing in it should be read as a system commissioned, adopted, or deployed by the DGAC.

Second, **no internal DGAC material was used at any stage.** The entire corpus is the publicly available EASA Easy Access Rules consolidation of Regulation (EU) No 376/2014 [5]; the benchmark questions are synthetic; no occurrence report, no database extract, and no internal document of the organisation enters the pipeline, the evaluation, or this document. The confidentiality question that normally attends an industry-hosted project therefore does not arise on the data side, and the transmission

of text to a hosted model, discussed in Section 4.11, concerns published regulatory material only.

3.2 The Regulatory Domain

The subject matter of the system is occurrence reporting in civil aviation, governed by Regulation (EU) No 376/2014 [4]. The source material used throughout this work is the EASA Easy Access Rules consolidation of that regime [5], which presents the base regulation together with its implementing and amending instruments and the associated guidance material in a single publication.

Understanding why this domain is difficult for an automated system requires understanding how the material is structured. The consolidated publication is not one document. As parsed and validated in Chapter 4, it comprises the four distinct legal instruments and the body of guidance set out in Table 3.1:

Table 3.1 – Composition of the source material, from the validated corpus.

Instrument	Art.	Ann.	Rec.	Subject
Regulation (EU) No 376/2014	24	3	1	The base occurrence-reporting regime
Regulation (EU) 2015/1018	2	5	1	Lists of occurrences subject to mandatory reporting
Regulation (EU) 2020/2034	4	1	1	Common European risk classification scheme
Regulation (EU) 2021/2082	6	1	1	Conversion from existing risk-classification tools
Guidance material	59 sections			Explanatory material on applying the regime

Three properties of this material shape every design decision in the project.

The instruments are separable and must stay separable. An article numbered 6 exists in more than one of these instruments. A retrieval system that treats the consolidated publication as a single document can return a provision from the wrong instrument while appearing to answer correctly, and a citation of the form “Article 6” is then ambiguous rather than wrong — which is worse, because ambiguity is harder to detect than error.

Binding text and guidance material are not interchangeable. Fifty-nine of the 109 legal units in the corpus are guidance material. Guidance explains how a requirement may be met; it does not itself impose the requirement. An answer that cites guidance where the regulation governs is superficially reasonable and legally wrong, and this distinction must therefore survive parsing, indexing, retrieval, and citation resolution rather than being recoverable only by a reader who already knows the difference.

The operative content is fine-grained. The base regulation’s articles address objectives, definitions, scope, mandatory reporting, voluntary reporting, collection and storage, quality and content of reports, the European Central Repository, exchange and dissemination of information, analysis and follow-up at national and Union level, confidentiality and appropriate use, protection of the information source, and penalties. Within any one of these, the obligation that answers a specific question typically sits in a single subparagraph, while the conditions that qualify it sit in neighbouring ones. The annexes of Regulation (EU) 2015/1018 are lists of reportable occurrences organised by operational domain — aircraft operation, technical condition and maintenance, air navigation services, aerodromes and ground services, and non-complex aircraft — and these are tabular and enumerative rather than prose.

3.3 Intended Users and Use

The system is intended to support a person who already has the competence to evaluate a regulatory answer but not the time to locate it: a safety officer determining whether an event is reportable, a compliance function checking a deadline or an addressee, an operations specialist confirming which annex covers a particular occurrence class.

That framing sets the acceptance criterion. The value of the system is not that it produces an answer; it is that it produces an answer whose supporting provision can be checked in seconds. An answer without a verifiable citation transfers the entire verification burden back to the user and therefore delivers no benefit. This is why the evaluation reported in this thesis measures citation support separately from answer correctness, and why abstention is treated as a legitimate output rather than a failure.

3.4 Mission and Its Revision

The initial mission produced three question-answering approaches over aviation regulations — a few-shot prompting baseline, a retrieval-grounded pipeline, and a LoRA-adapted model — together with stored model predictions, question sets, source chunks, evaluation outputs, and an intermediate report. The initial research question asked which of the three performed best.

A repository and metric audit showed that this comparison could not isolate the causes of failure, for the reasons set out in Section 1.2. The revised mission preserves the three-system comparison as a historical baseline and adds the following deliverables:

- an audited and hierarchy-aware regulatory corpus with preserved instrument identity;

- an evidence-linked evaluation benchmark with source-grouped splits and a documented validity gate;
- controlled retrieval, fusion, and reranking ablations under a fixed corpus and split policy;
- an evidence-constrained generation and verification protocol;
- a failure taxonomy and an escalation policy driven by observable runtime signals;
- a prototype that displays evidence, citations, abstention state, and review status;
- a reproducible research package and the final report.

The revision was recorded as a dated decision rather than applied silently, and the earlier artefacts were preserved unchanged rather than overwritten, so that the two phases of the project can be compared as they actually were.

3.5 Data and Provenance

Three source artefacts are recorded in the project manifest with their SHA-256 digests and marked immutable: the Easy Access Rules XML export used as the primary corpus, the consolidated regulation text as the authoritative reference, and the Easy Access Rules PDF as a rendering reference for manual checks. Only the XML export is parsed; the other two support human verification of the parser’s output.

From this source the project derives 109 coherent legal parents and 1,535 retrievable children. Each child records its parent, hierarchy path, canonical citation, unit type, sequence, source range, and an indexing text distinct from its displayed evidence text. The rationale for that separation, and the validation applied to the result, are given in Chapter 4.

The historical evaluation material consists of 300 synthetic question–answer records generated over the earlier flat-chunk corpus. A 100-question subset is shared by the stored few-shot, RAG, and LoRA result files, and it is this subset that carries continuity with the baseline. After the evidence audit, 99 records remain in the candidate benchmark.

The provenance of this benchmark must be stated plainly, because it bounds what the thesis can claim. The questions were machine-generated. The reference answers were machine-generated. The evidence links were proposed with machine assistance and then corrected. It is a research instrument, not an independently authored legal examination set, and the validity gate described in Chapter 4 exists precisely because that provenance cannot be argued away.

3.6 Constraints

The project operates under six constraints.

1. **Reliability.** A fluent answer without exact supporting evidence is not sufficient for aviation-regulation use. Citation support is a first-class requirement, not a presentational feature.
2. **Benchmark validity.** Synthetic questions and machine-assisted corrections require transparent disclosure and independent review before any figure is presented as final.
3. **Leakage control.** Questions grounded in the same legal parent cannot be split across development and test, and no threshold may be fitted on the test partition.
4. **Compute and cost.** Hosted or expensive methods must be justified by measured gains, and their inputs, token use, latency, and cost recorded. Where a local model suffices, it is preferred — as with reranking, which runs entirely on local hardware.
5. **Data export.** The regulatory sources are public, but transmitting project questions and passages to a hosted service is treated as an explicit decision requiring authorisation, not a routine operation.
6. **Time.** The benchmark, end-to-end evaluation, prototype, report, and defence must be completed within the submission calendar, which is compressed relative to the original plan.

The fifth constraint has already shaped the project’s history. An initial attempt to compute dense embeddings was blocked because it would have exported corpus and question text without explicit authorisation; the experiment was suspended, the authorisation obtained and recorded, and only then were the two embedding runs executed, with the exact token counts logged. Final hosted answer generation is subject to the same requirement and remains unauthorised at the time of writing.

3.7 Infrastructure

Corpus construction, lexical retrieval, evaluation, context assembly, and verification run locally in Python. Dense vectors are cached by model identifier and input hash, so that a change to a label does not force re-embedding when the underlying text is unchanged. The cross-encoder reranker runs locally on an NVIDIA RTX 4070 Laptop

GPU, and its 1,980 query–passage scores are cached together with the model revision and the experiment parameters.

This arrangement serves two purposes beyond cost. It keeps project text out of hosted inference at the reranking stage entirely. And it makes re-scoring against corrected labels cheap: when the benchmark is frozen with reviewer corrections applied, the retrieval and reranking outputs do not need to be recomputed, only re-evaluated.

3.8 Governance and Traceability

All new research work is isolated in a versioned project directory; historical artefacts remain unchanged. Every promoted corpus, benchmark, ranking, context, and report asset carries deterministic identifiers or SHA-256 lineage, so that any result can be traced to the exact inputs that produced it. Experiment directories are immutable by default, and each manifest records parameters, software and model versions, hardware, timestamps, and a status field.

Architectural decisions are recorded in an append-only decision log, with reversals entered as new decisions referencing the ones they supersede. Rejected ablations remain documented. Artefacts produced with machine assistance are recorded in a separate log that also states what verification was performed on each and what author work remains outstanding; that log is the source of record for the disclosure in Appendix B.

3.9 Deliverable Schedule

The schedule below reflects the revision of 25 July 2026, which moved the full-draft target forward to accommodate a supervisor review before the end-of-August submission.

25 July – 6 August	Multi-reviewer collection on the benchmark validity gate, in parallel with author adjudication of the review queue.
6 – 7 August	Response aggregation, adjudication of contested records, and benchmark freeze.
7 – 10 August	End-to-end generation, structural and semantic verification, and abstention.
10 – 11 August	Escalation calibration on development data, held-out reporting, final tables and figures.
11 – 12 August	Completion of the results and conclusion chapters; full draft circulated.

13 – 25 August Supervisor feedback incorporated; discussion, limitations, ethics, and conclusion finalised. No new major experiments.

26 – 31 August Final revision, reference and caption checks, traceability verification, and submission.

One scheduling rule is worth stating explicitly because it is a methodological commitment rather than a convenience. The review collection closes on 6 August regardless of the coverage achieved, and the benchmark is frozen on whatever coverage exists at that date. A cutoff that moves whenever a response is outstanding is not a schedule; more importantly, a cutoff chosen after observing the response rate would allow coverage to be selected in favour of a preferred result. Records that have not reached the two-reviewer minimum at the cutoff fall to author adjudication and are reported as such, and any response arriving after the freeze is retained and reported as a robustness check rather than merged into the frozen set.

3.10 Risks and Mitigations

Table 3.2 lists the principal risks identified at the outset of the project and the mitigation applied to each.

The last row describes what actually happened. The state of the art, methodology, introduction, and project context were written during the period in which generation was blocked behind the benchmark freeze, which converted a scheduling constraint into writing time rather than into idle time.

Table 3.2 – Principal project risks and the mitigation applied to each.

Risk	Mitigation
Insufficient reviewer coverage by the cutoff	The author fills the review queue in parallel; records below the minimum are adjudicated and reported separately rather than dropped.
Corpus defects mistaken for model limitations	Corpus repaired and deterministically validated before any retrieval error is attributed to a model.
Benchmark provenance undermining final claims	Stratified human audit with reported agreement; both the immutable baseline view and the audited view reported.
Evidence leakage inflating results	Parent-grouped partitioning with a programmatic assertion that no group spans both splits.
Hosted generation not authorised in time	Generation is the last stage and is isolated; the retrieval and reranking contributions stand independently of it.
Scope creep into multi-step retrieval	Multi-step retrieval is admitted only for a residual failure shown, after review, to require it.
Report written too late	Chapters not dependent on final results were drafted while experiments were still blocked.

Chapter 4

Methodology

This chapter specifies the system the study measures, in the order in which a question passes through it. Figure 4.1 shows that path in full; the sections that follow take it one component at a time, and each records what was fixed on development data and what was left open.

4.1 Research Design

4.1.1 Design Principles

The study is a staged ablation. Each stage changes one major component while holding the corpus, the benchmark identifiers, and the source-grouped partition fixed, so that a change in measured performance can be attributed to the component that moved. Four principles govern the design and are enforced in code rather than by convention.

1. **Retrieval is evaluated before generation.** A failed answer can originate in missing evidence, poor ranking, context construction, generation, or verification. Aggregating these into one score destroys the information needed to choose an intervention, which is the object of study here.
2. **One component changes at a time.** Where two changes are of interest, they are run as separate experiments with separate manifests rather than as a combined configuration.
3. **Selection happens on development data only.** Model choice, fusion constants, and any threshold are fixed on the development partition. The test partition is used for held-out reporting and never for tuning.
4. **Failed and rejected configurations are retained.** Negative ablations remain in the repository and in the decision log rather than being silently replaced, so that the reported configuration can be read against the alternatives that were tried.

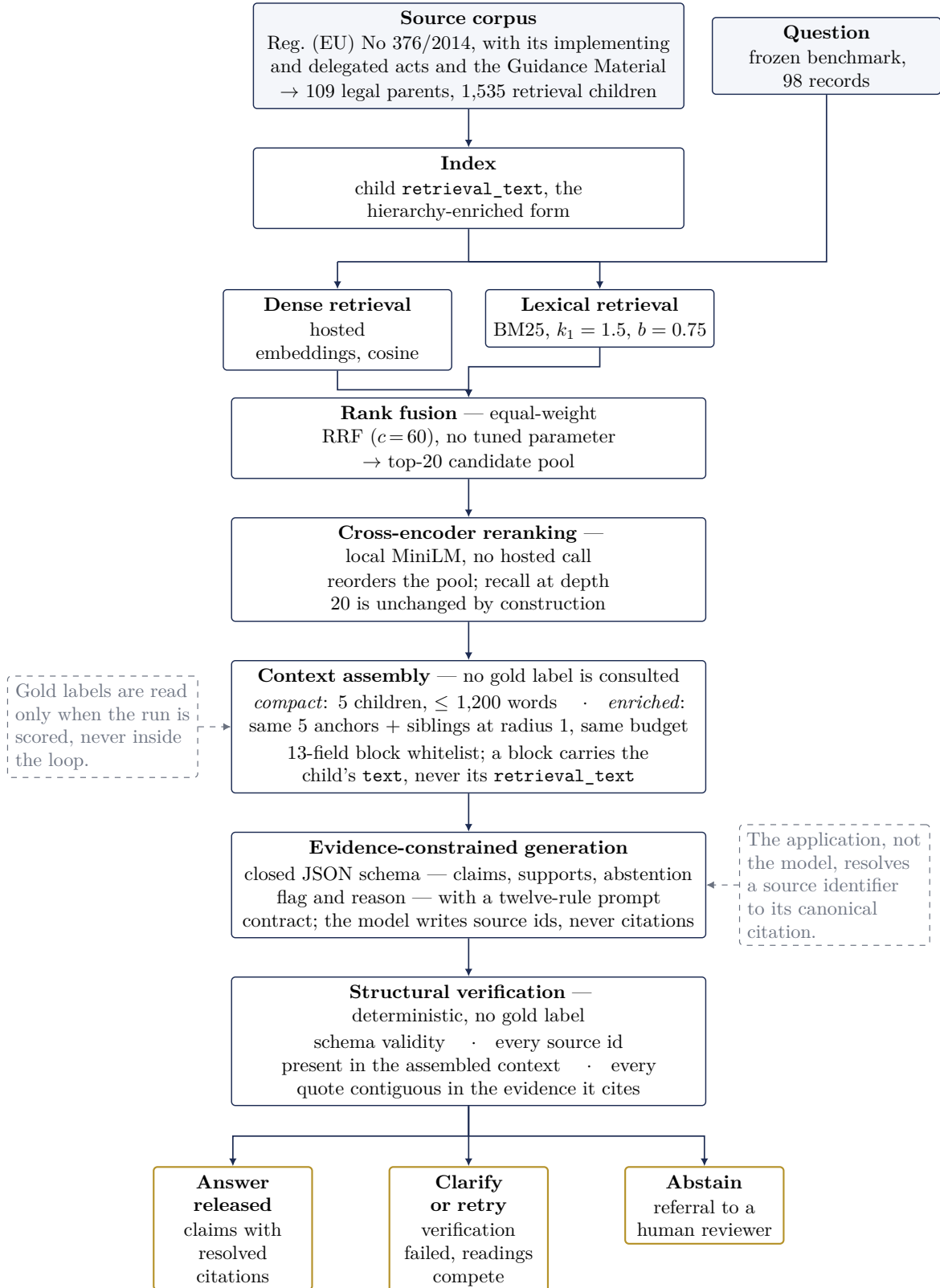


Figure 4.1 – The evaluated pipeline, end to end. Solid arrows carry the runtime path; dashed boxes record two properties that hold throughout. The generator is given source identifiers and returns them, so a citation string is produced by the application and never by the model; and no gold label is read at any point between the question arriving and an outcome being chosen. Retrieval parameters are those fixed on the development partition (Section 4.4).

Listing 4.1 shows the third principle as it exists in the code. The generation driver defaults to the development partition and refuses the held-out one outright unless a flag is passed whose name is a statement the author has to be willing to make. A convention can be forgotten under deadline; this cannot.

```

1     if args.split in ("test", "all") and not
        args.i_have_frozen_calibration:
2         raise SystemExit(
3             f"Refusing to generate on '{args.split}': the held-out
                split is only "
4             "scored once, after calibration is frozen on
                development data. "
5             "Re-run with --i-have-frozen-calibration if that is
                genuinely done."
6         )

```

Listing 4.1 – The held-out split cannot be generated by accident. Source: `scripts/run_generation.py`, lines 181–186.

4.1.2 Experiment Ladder

The ladder proceeds from the frozen historical baseline to the interventions under study. Each rung isolates a single decision.

The stages are set out in Table 4.1.

Table 4.1 – Experiment ladder. Each stage alters one component relative to the stage above it.

ID	Chunking	Retrieval	Component under test
E0	Flat legacy	Dense (3-small)	Frozen historical baseline
E1	Parent–child	Dense (3-small)	Effect of hierarchy-aware chunking
E2	Parent–child	Dense (3-large)	Effect of embedding capacity
E3	Parent–child	BM25 + dense, RRF	Effect of a second retrieval channel
E4	Parent–child	E3 + cross-encoder	Effect of reranking a fixed pool
E5	Best configuration	E4	Context assembly, generation, verification
E6	Best configuration	E4	Optional constrained multi-step retrieval

Operationally the sequence is: audit the historical corpus and evaluation artefacts; construct a deterministic parent–child corpus; map historical questions to exact legal evidence; compare two dense embedding models on identical inputs; add local BM25 and equal-weight RRF; rerank the fused top-20 pool with a local cross-encoder; assemble bounded evidence contexts; generate structured answers and verify claims, quotes, and citations; and calibrate clarification, retrieval expansion, abstention, and human-review triggers.

The experiment registry was reconciled with the stored manifests on 25 July 2026. It records E1–E4 as completed candidate runs, resolves E3 to `text-embedding-3-large` under D014, and distinguishes completed E5 context assembly from the generation phase, which was still blocked at that date and has since been run (Chapter 5).

4.1.3 Unit of Analysis

The unit of analysis is the question. For each question the study records the retrieved candidate set at several depths, the reranked ordering, the assembled context, the generated structured answer, the outcome of each verification check, and the escalation level at which the question was resolved. Aggregate figures are computed over questions within a partition; no figure is reported without its denominator.

4.2 Corpus Construction

4.2.1 Defects in the Historical Corpus

The historical corpus contained 132 comparatively flat chunks derived from the same regulatory source. Three defect classes were identified in the audit and had to be repaired before any retrieval error could be interpreted as a model limitation.

First, chunk size was uncontrolled: some chunks exceeded 30,000 characters while the generator received only the first 1,500 characters of the selected chunk. A source could therefore be counted as retrieved while the supporting paragraph was absent from the generation context, which makes retrieval success and answer success only loosely coupled. Second, document identity was not preserved: the Easy Access Rules publication consolidates several instruments, and similarly numbered provisions from different instruments could collide. Third, legal type was not reliably separated, so binding regulation text and Guidance Material (GM) were at risk of being treated as equivalent evidence.

The third defect is the most consequential for this application. Guidance material explains how a requirement may be met; it does not itself impose the obligation. A system that cites guidance where the regulation governs produces an answer that is superficially reasonable and legally wrong.

4.2.2 Source Snapshot and Provenance

Three source artefacts are recorded in `data/manifests/source_manifest.csv` with their SHA-256 digests and marked `immutable_historical`: the EASA Easy Access Rules XML export used as the primary corpus, the consolidated regulation text as the authoritative reference, and the Easy Access Rules PDF as a rendering reference for

manual checks [4], [5]. Only the XML export is parsed; the other two support manual verification.

The build manifest for the promoted corpus records the source path, the source digest, the schema version `parent_child_v1`, the experiment identifier, start and completion timestamps, the validation summary, and the digests of the emitted parent and child files. Downstream experiment manifests inherit those digests, so any change to the corpus invalidates the lineage of every result computed from it.

4.2.3 Legal Identity

The parser preserves the identity of each underlying instrument rather than treating the consolidated publication as one document. The promoted corpus spans five document identities: the base regulation, three amending or implementing instruments, and the guidance material issued for the base regulation. Each legal parent records its document, regulation, type, and version; its formal title and legal-unit type; its hierarchy path, parent title, and canonical citation; its cross-references and source range; and the ordered identifiers of its retrievable children.

This identity layer is what prevents an article numbered 6 in one instrument from being confused with an article numbered 6 in another, and it is what keeps regulation and guidance distinguishable at retrieval time rather than only at reading time.

4.2.4 Parent–Child Representation

A parent is a coherent legal unit: a complete article, an annex section, a recitals block, or a guidance section. A child is an atomic retrievable unit: a paragraph, list item, table row, table header, or callout. The design follows the finding that finer indexing granularity improves retrieval, particularly for rare entities [28], while retaining the structural context that makes a fragment of legal text interpretable.

Table 4.2 lists the fields each unit stores and which of them are optional.

Table 4.2 – Stored fields. Parents carry fourteen fields, of which only cross-references are optional; all eleven child fields are required.

Record	Fields
Parent	<code>parent_id</code> , <code>document_id</code> , <code>document_version</code> , <code>regulation_id</code> , <code>document_type</code> , <code>hierarchy_path</code> , <code>canonical_citation</code> , <code>title</code> , <code>text</code> , <code>source_location</code> , <code>cross_references</code> (optional), <code>char_count</code> , <code>word_count</code> , <code>child_ids</code>
Child	<code>child_id</code> , <code>parent_id</code> , <code>sequence</code> , <code>canonical_citation</code> , <code>unit_type</code> , <code>text</code> , <code>retrieval_text</code> , <code>char_count</code> , <code>word_count</code> , <code>marker_path</code> , <code>source_location</code>

Two field distinctions carry methodological weight. The child stores both `text` and `retrieval_text`: the former is the evidence exactly as it appears in the source and is

what a verifier and a human reader see, while the latter is a metadata-enriched form used for indexing. Separating them means that hierarchy can improve retrieval without contaminating the evidence that a quote is checked against. Table rows repeat the governing table and column headers in `retrieval_text` for the same reason: the row becomes findable without the displayed evidence being altered.

Short legal units are not merged to satisfy a generic minimum length. An atomic item remains interpretable through its stored hierarchy, and merging would reintroduce exactly the boundary confounding that the repair was intended to remove (decision D007).

4.2.5 Deterministic Validation

Validation is implemented as fifteen distinct checks — eleven critical and four warning — applied at corpus, parent, child, and cross-reference level. Critical checks reject an empty corpus or empty child set, duplicate parent or child identifiers, empty parent or child text, empty retrieval text, a missing regulation identifier, a parent with no children, an orphaned child, an article whose title does not begin with “Article”, and any mismatch between a parent’s declared child ordering and the children that actually reference it. Warning checks flag an unusual guidance title, a parent above 40,000 characters, a child above 4,000 characters, and a table row that carries no column structure while exceeding twenty words.

The child-order check is the one that most directly protects downstream work: because context assembly expands to neighbouring siblings by position, an ordering mismatch would silently return the wrong neighbours.

Table 4.3 reports the composition of the corpus that passed validation.

Table 4.3 – Validated E1 corpus composition, from the stored validation record.

Unit	Count
Legal parents	109
Retrievable children	1,535
ARTICLE parents	36
GM parents	59
ANNEX parents	10
RECITALS parents	4
List-item children	823
Paragraph children	477
Callout children	122
Table-row children	105
Table-header children	8

The promoted corpus reports zero critical issues and zero warnings. Child length

ranges from 2 to 1,615 characters with a mean of 192.61. The corpus audit additionally reports 87 units below 20 characters and 8 units above 1,000 characters; these two counts appear in the audit report only and are not recorded in the validation output. A second independent build produced identical parent, child, and validation bytes, and the builder refuses to overwrite a non-empty output directory unless forced.

4.3 Benchmark Construction

4.3.1 Legacy Lineage

The historical dataset contains 300 synthetic question–answer records generated over the 132 flat chunks. Because the old source field records a chunk rather than a legal unit, it cannot be treated as exact evidence. Every record was therefore mapped into the E1 hierarchy under a four-stage precedence: normalised text equality; legacy text wholly contained in a new parent; a single new parent wholly contained in a legacy chunk; and, last, fuzzy token coverage with cosine ranking. Crucially, the fourth stage never accepts a mapping — it marks the record as requiring review.

Table 4.4 reports the outcome of that mapping over the 300 historical records.

Table 4.4 – Legacy parent-lineage audit over 300 historical records.

Disposition	Records
Automatically mapped	291
Manually accepted	4
Excluded as invalid	4
Rewrite required	1
Total	300

At chunk level, 113 of the 132 legacy chunks mapped automatically and 19 required review; of those 19, only 4 were used by question records lacking an automatic mapping, and the 9 affected records were resolved individually and recorded with their resolutions. This asymmetry matters: the number of ambiguous chunks overstates the number of ambiguous evaluation records by a factor of roughly two.

4.3.2 Evidence Audit of the Shared Evaluation Set

The 100 question identifiers shared by the stored few-shot, RAG, and LoRA outputs were then linked to exact child evidence. This audit was Artificial Intelligence (AI)-assisted, and the stored decision counts record exactly what was done to each record: 78 retained without text change, 21 rewritten and evidence-linked, and 1 excluded as

invalid. The retained candidate benchmark contains 99 records, each identifying a document-qualified parent and at least one selected evidence child.

The excluded record, QA_0003, was generated from concatenated table-of-contents text; its reference answer merely refers the reader to a section number and is not evidence-grounded. All three baseline systems failed its citation criterion, so its removal does not favour any system. Because the historical results were computed over 100 questions, both views are reported: the immutable $N = 100$ baseline for continuity, and the $N = 99$ sensitivity view for any claim that depends on evidence-linked gold data (decision D009).

4.3.3 Independent Multi-Reviewer Validation

The audit described above cannot certify itself. The same AI-assisted process that proposed corrections is not in a position to confirm them, and the wider literature gives direct grounds for caution: model-generated evaluation data requires human anchoring before its scores can be trusted [58], and model judges carry documented position, verbosity, and self-enhancement biases [59]. A separate validity gate is therefore mandatory before any figure computed on this benchmark is treated as final.

The review package contains 33 records selected deterministically: all 22 corrected or excluded audit exceptions; all three hybrid top-20 residuals, two of which already appear among the exceptions; and ten unchanged controls stratified by source class and split at two per stratum. The sample spans 16 ARTICLE, 13 GM, and 4 ANNEX records, and 16 development, 16 test, and 1 excluded record. The required identifiers are pinned in a requirements manifest together with the digests of the audited benchmark, the child corpus, and the split manifest, so that the review cannot be applied to a different version of the data than the one it was performed on.

For each record a reviewer records question clarity, answer support, citation correctness, evidence completeness, corrections, exclusions, and alternative acceptable evidence sets. Multi-child alternatives are represented as logical evidence routes rather than as a single union that must always be retrieved together, so that a system retrieving a legally sufficient alternative is not scored as having failed (decision D021).

The review is conducted by a panel of practising aviation professionals rather than by the author alone (decisions D023 and D024). Three properties of the protocol are fixed before any response is collected. First, a record is auto-decided only when at least two reviewers have rated it. Second, the aggregation rule is a strict majority on the overall decision and on each of the four validity checks, with **accept** requiring all four checks to resolve affirmatively. Third, a majority verdict of “accept with correction” is never applied automatically: the panel identifies the problem and the author supplies the corrected fields, because a corrected evidence set cannot be produced

by a multiple-choice instrument.

Agreement is reported rather than asserted, using chance-corrected statistics: Cohen’s coefficient for two raters [61], Fleiss’s generalisation for three or more [62], with magnitudes interpreted against the conventional benchmarks and their arbitrariness acknowledged [63]. Records on which qualified reviewers disagree are reported individually rather than resolved by majority and absorbed into the aggregate. Disagreement between practitioners about whether a provision supports an answer is evidence that the question is ambiguous, and treating it as noise would discard the signal that identifies precisely the cases the escalation policy exists to handle.

The freeze command validates the exact 33 required identifiers, reviewer role and date, allowed decision values, evidence identifiers, source digests, and correction consistency. It then applies corrections, preserves provenance, excludes rejected records transparently, regenerates the parent-grouped splits, and writes SHA-256 manifests. It refuses to run against an incomplete review and refuses to overwrite a non-empty output directory.

4.3.4 Parent-Grouped Partitioning

Random splitting at question level would leak near-identical evidence whenever several questions originate from one article or guidance section, and such leakage inflates apparent generalisation. The benchmark is therefore partitioned by legal parent, following the practice of reporting evidence provenance alongside answers [54].

The partitioner is deterministic without a random number generator. Groups are ordered by descending size, with ties broken by the SHA-256 digest of a string seed concatenated with the group identifier, and are then assigned greedily to whichever partition brings the running development count closest to a target fraction of records. Because the seed is a string consumed by a hash rather than an RNG state, the split reproduces exactly across platforms and library versions. A separate assertion raises an error if any group’s questions carry more than one partition label.

The resulting partition is given in Table 4.5.

Table 4.5 – Candidate parent-grouped evaluation split, from the stored split summary.

Partition	Questions	Parent groups
Development	42	12
Test	57	48
Total	99	60

Two features of Table 4.5 require comment, since both look like errors and are not. The requested development fraction is 0.4, which for 99 records implies a target of 40; the greedy rule lands on 42 because groups are assigned whole and the target cannot generally be hit exactly. The development partition holds 42 questions across only

12 parents while the test partition holds 57 across 48, because the largest groups are ordered first and are therefore assigned first. Recorded group leakage is zero.

The split and the review sample use deterministic string seeds:

- `aviation-rag-legacy100-v1` for the parent-grouped partition;
- `aviation-rag-independent-review-v1` for the review sample.

Both are consumed by SHA-256 assignment; no integer random-number-generator seed is used for these operations.

4.4 Retrieval Architecture

4.4.1 Dense Retrieval

Two hosted embedding models process identical inputs — the 1,535 hierarchy-enriched child retrieval texts and the 99 candidate questions. Each model-specific run consumed 172,821 input tokens, for 345,642 across both. Vectors are cached with their input digests, model names, and request metadata, so that a change to a label does not force re-embedding when the underlying text is unchanged. Cosine similarity ranks child vectors for each query.

The larger model was selected on the development partition only, before the test comparison was opened (decision D014). No in-domain fine-tuning of the embedding model was performed; the design relies on the finding that general-purpose contrastively trained retrievers transfer to unseen domains without task-specific training [21].

4.4.2 Lexical Retrieval

BM25 is implemented locally with $k_1 = 1.5$ and $b = 0.75$. For query Q and child document D ,

$$\text{BM25}(D, Q) = \sum_{q \in Q} \text{IDF}(q) \frac{f(q, D)(k_1 + 1)}{f(q, D) + k_1 \left(1 - b + b \frac{|D|}{\text{avgdl}}\right)}, \quad (4.1)$$

where $f(q, D)$ is the term frequency, $|D|$ is the document length in tokens, and avgdl is the corpus mean. Inverse document frequency uses the non-negative form $\text{IDF}(q) = \log\left(1 + \frac{N - n_q + 0.5}{n_q + 0.5}\right)$, and each term's contribution is additionally scaled by its query frequency; the formulation follows the standard treatment of term weighting and inverted indexing [65]. Ties are broken by corpus position, so rankings are deterministic. The channel defined by Equation 4.1 supplies exact-terminology matching for article references, acronyms, deadlines, and repeated regulatory expressions [17].

Tokenisation is deliberately minimal: text is lower-cased and split on the pattern `[a-z0-9]+`, with no stopword removal, stemming, or lemmatisation. Two consequences must be stated plainly rather than discovered by a reader. A citation such as “376/2014” tokenises into two separate numeric tokens, and a subparagraph reference such as “4(1)(a)” into four; the lexical channel therefore matches the components of a legal reference rather than the reference as a unit. And because the character class is restricted to unaccented ASCII after lower-casing, accented characters act as token separators, which bounds what this implementation could do on French-language queries without modification. Learned sparse retrieval would address the first limitation by design [18]; it is recorded here as future work rather than as an implemented alternative.

4.4.3 Rank Fusion

The selected dense ranking and the BM25 ranking are combined by equal-weight RRF, using Equation 2.1 with $c = 60$. Ranks are one-based, duplicates within a single input ranking are ignored, and ties are resolved first by the best rank achieved in any input and then lexicographically by identifier, so that the fused ordering is reproducible across platforms. The constant was fixed before held-out reporting and was not tuned on test data.

Two alternatives were evaluated on development data and rejected: unequal fusion weights, and boosting candidates by parent mass. Neither improved the top-20 candidate objective, and parent boosting consistently reduced recall. Both are retained as documented negative ablations (decision D017).

The choice of RRF over a tuned score-level combination is a deliberate trade of performance for methodological safety. A convex combination of normalised scores has been shown to outperform RRF both in and out of domain, and rank-based fusion necessarily discards score magnitude [27]. A fusion function with no tuned parameters, however, cannot leak information from the held-out partition, which in a study whose central claim concerns measurement validity is worth more than the marginal recall forgone. The trade-off is revisited in the limitations.

4.4.4 Cross-Encoder Reranking

The fused top-20 candidates are reranked by a cross-encoder that scores each query–passage pair jointly, following the established result that direct query–passage interaction substantially improves ranking over independently encoded representations [30]. The model is a distilled English passage-ranking cross-encoder [32] trained on the MS MARCO passage collection [33], run with a maximum sequence length of 512 and a batch size of 64 on a local CUDA device. The run scored 1,980 query–passage pairs over 99 queries, and the manifest records the model identifier, the exact model revision

commit, the framework version, the device, and the digest of the cached score set. No question or regulatory passage was sent to a hosted inference endpoint during reranking.

The training distribution of this reranker must be named rather than assumed away. MS MARCO consists of short, factoid-oriented web search queries; the questions in this benchmark are longer, structurally explicit, and drawn from a specialist regulatory sublanguage. Any improvement is therefore obtained despite a domain shift, and any residual failure has to be read with that shift in view. A multilingual reranker was considered and deferred on scope and model-size grounds (decision D018).

The reranking implementation preserves each candidate’s original hybrid rank alongside its new score, so that recovery and regression relative to the pre-reranking ordering can be measured per question rather than only in aggregate. A reranker can repair ordering only when the relevant evidence is already present in the candidate pool; recall at depth 20 is therefore expected to be unchanged, and that invariance is what distinguishes a ranking failure from a candidate-generation failure.

4.5 Context Construction and the Generation Contract

4.5.1 Context Selection

Two deterministic context variants are assembled from the reranked ordering, neither of which consults gold labels at any point. The compact variant selects the top five children under a 1,200-word budget with no neighbour expansion. The enriched variant selects the same five anchors, expands to adjacent sibling units within a radius of one, and applies the same 1,200-word budget. An earlier build of the variant, over the 99 pre-review questions, used a 1,600-word budget instead; the difference makes no selection difference on this corpus, for the reason given below. Budgets are measured in stored child word counts.

Three behaviours of the selection procedure should be understood precisely, because each would otherwise look like a defect. The first anchor is admitted unconditionally, even if it alone exceeds the budget, so that no question is ever left with an empty context. A candidate skipped for budget does not stop the scan, so a lower-ranked but shorter child may still be admitted — selection is best-fit by rank, not a prefix of the ranking. And neighbours receive no budget exemption and are drawn strictly from siblings within the same parent, never across parents, so expansion cannot silently introduce a different legal unit.

Bounding the context is not a resource optimisation. Models use evidence unreliably when it sits in the middle of a long context, and performance degrades as context length

grows even in models designed for long inputs [29]. A tight, ordered evidence budget is therefore a correctness measure, and it is what makes reranking worth its cost.

Each context block exposes a fixed whitelist of thirteen fields: source identifier, parent identifier, role, anchor identifier, retrieval rank, sequence, canonical citation, document identifier, document type, document version, hierarchy path, unit type, evidence text, and word count. Notably the block carries the child’s `text`, not its `retrieval_text`; the metadata-enriched indexing form is never shown to the generator, so a quote can be checked against exactly what the model was given. Parent title and text, source locations, and marker paths are deliberately withheld.

Table 4.6 reports what the selection procedure actually produces on this corpus.

Table 4.6 – Observed context statistics for the enriched variant, as built over the 98 frozen questions — the artefact that generation reads. The earlier build over the 99 pre-review questions gave 6.36 neighbours and 473.3 evidence words per question under the same selection rule.

Quantity	Value
Anchors per question (mean)	5.00
Neighbours per question (mean)	6.27
Sources per question (mean / median / max)	11.27 / 11 / 15
Evidence words per question (mean / median / max)	463.8 / 451.5 / 1,090
Questions hitting the word budget	0

The maximum of fifteen sources is the structural ceiling for five anchors at radius one. No candidate was skipped for budget, and the largest observed context is 1,090 words against a 1,200-word ceiling. Because nothing was skipped, a larger ceiling could not have admitted any further unit: the enriched variant is bounded by structure rather than by the budget in this corpus, which is why the earlier 1,600-word setting and the present one select the same evidence.

4.5.2 Answer Schema

The generator returns a single JSON object validated against a closed schema that forbids additional properties. Five top-level properties are required: the answer text; an array of claims; an abstention flag; an abstention reason, which is either null or one of five controlled values (ambiguous question, conflicting sources, insufficient evidence, outside corpus, verification failed); and a clarification question, which may be null.

Each claim carries a non-empty identifier, its text, and at least one support. Each support carries a non-empty source identifier and a non-empty verbatim quote. The minimum-item constraint on supports is the structural core of the contract: a claim with no support cannot validate.

Two limits of the schema are recorded here rather than left for a reader to find.

It does not encode the cross-field rule that the abstention reason must be null when the abstention flag is false — that rule is stated in the prompt and must be checked separately. And it does not restrict source identifiers to the supplied set; that restriction is applied at runtime against the identifiers actually present in the assembled context.

4.5.3 Prompt Contract

The prompt instructs the model to answer only from the supplied source blocks, not to rely on unstated legal knowledge, and not to invent article numbers, citations, obligations, exceptions, deadlines, or consequences. Twelve numbered rules operationalise this: decompose the answer into the smallest material claims; give each claim a unique identifier; support every claim with one or more supplied source identifiers; attach a short verbatim quote to each support; never write an unsupplied source identifier; do not generate display citations at all; abstain with the appropriate controlled reason when evidence is insufficient, when the question admits several plausible legal meanings, when supplied sources materially conflict, or when the question falls outside the corpus; set the abstention reason to null when not abstaining; and distinguish binding regulation text from guidance where the distinction affects the answer.

The sixth rule carries the most weight. The application, not the model, resolves source identifiers to canonical citations, so a model cannot produce a valid-looking citation string by itself. This follows the finding that retrieve-then-read pipelines outperform post-hoc citation attachment on both correctness and attribution [44], and it directly targets the failure mode measured in deployed systems, where citations are frequently present but do not support the statements they are attached to [2].

4.6 Verification and Selective Answering

4.6.1 Structural Verification

Structural verification is deterministic and applies three checks. Schema validation rejects malformed output. The allowed-source check rejects any source identifier not present in the assembled context. The quote check requires each supporting quote to occur in the evidence text of the source it cites, compared after collapsing whitespace, trimming, and case folding. Punctuation and hyphenation are not normalised, so the check is strict on exactly the features that distinguish one legal formulation from another.

The outcome is a decision among three states: structurally accepted, abstained, or returned for retry or abstention. One asymmetry in the implementation should be understood by anyone reading the results. A claim can be marked unsupported —

because its source identifier is unknown or its quote is blank — without contributing an entry to the quote-error list. The decision state can therefore be “structurally accepted” while a specific claim is not among the supported claims. Support coverage must accordingly be read from the list of supported claim identifiers rather than inferred from the decision.

Listing 4.2 is that quote check. Normalisation removes typography and punctuation, which the corpus inherits from a PDF extraction and which carry no legal meaning; the containment test is then applied to the normalised word sequence, so a quote is accepted only where its words appear *contiguously* in the evidence. Contiguity is not relaxed, and it is the condition that rejects a quote assembled from separated fragments.

```

1 def _normalize_for_matching(value: str) -> str:
2     """Typography- and punctuation-insensitive; word order is
3         preserved."""
4     value = NON_WORD.sub(" ", _normalize_typography(value))
5     return WHITESPACE.sub(" ", value).strip().casefold()
6
7 def quote_is_grounding(quote: str, evidence_text: str) -> bool:
8     """Does the quote's word sequence appear contiguously in the
9         evidence?"""
10    return _normalize_for_matching(quote) in
11           _normalize_for_matching(evidence_text)

```

Listing 4.2 – The check the argument of this thesis rests on. Source:

src/aviation_rag/generation/verification.py, lines 72–80; reproduced
with the byte-level rule beside it in Appendix D.

4.6.2 The Limits of Structural Verification

Structural checks establish provenance, not entailment. A quote may be reproduced perfectly and still fail to support the claim built on it, and material omission — an answer that states an obligation while omitting the condition that suspends it — is invisible to a containment test. This is the extrinsic case that dominates the risk profile in generation over authoritative text [41], and the empirical record in deployed legal systems shows that retrieval grounding reduces but does not eliminate it [1].

Accordingly, every non-abstaining answer is flagged for separate semantic review, and claim support and completeness are evaluated independently of the structural pass rate. The formal criterion for that review is attributability: a claim counts as supported only if a competent reader would affirm that, according to the cited source, the claim holds [43]. Reporting a structural pass rate as though it were a groundedness rate would repeat precisely the conflation this thesis sets out to measure.

4.6.3 Selective Answering

The system records accepted answers, retries, clarification requests, and abstentions, and selective performance is reported as a risk–coverage relationship rather than as a single accuracy figure [47]. The policy prefers a correct refusal to an unsupported regulatory answer, and this preference is a design commitment rather than an empirical finding.

The signal driving abstention is deliberately not the generator’s own confidence. Model confidence is poorly calibrated under domain shift, and a decision function separate from the answering model outperforms thresholding the model’s own score [48]. The triggers used here are therefore external and observable: retrieval score margin, disagreement between the lexical and dense channels, absence of any candidate above threshold, and the outcome of structural verification.

4.7 Escalation Policy and Its Calibration

The escalation policy is the object of study rather than a component of the system, and it is specified as a ladder of levels with an explicit transition rule between them. Table 4.7 sets out the levels and the condition for moving beyond each of them. The same ladder is redrawn in Figure 5.3, in Chapter 5, once the results have fixed the evidential status of each transition.

Table 4.7 – Escalation levels and their trigger conditions. All thresholds are calibrated on development data only.

Level	Action	Trigger for moving beyond this level
0	Dense retrieval only	Baseline; superseded by construction
1	Hierarchy-aware retrieval	Default after corpus repair
2	Hybrid candidates with fusion	Low fused score, or lexical/dense disagreement
3	Cross-encoder reranking	Small margin between top candidates
4	Evidence-constrained generation	Applied to all answered questions
5	Clarification or targeted revision	Verification failure, or several plausible readings
6	Abstention and human referral	Conflicting sources, no supporting evidence, unresolved ambiguity

Three constraints govern calibration. First, every trigger must be computable at runtime: gold-label agreement is unavailable in deployment and cannot appear in a trigger, however convenient it would be for reporting. Second, thresholds are fitted on the development partition and applied unchanged to the test partition. Third, the policy is evaluated not only on final accuracy but on whether it escalated the right questions, through escalation precision — the proportion of escalated questions that would in fact have failed at the lower level — and escalation recall — the proportion

of failing questions that were escalated. A policy that escalates everything achieves perfect recall and is worthless; reporting both quantities is what makes the claim of an economical policy falsifiable.

4.8 Evaluation Measures

4.8.1 Retrieval Measures

For a question j with a set of acceptable evidence routes $\mathcal{G}_j$ and retrieved top- k set $R_{j,k}$, evidence-route recall is

$$\text{EvidenceRecall}_j @k = \max_{G \in \mathcal{G}_j} \frac{|R_{j,k} \cap G|}{|G|}. \quad (4.2)$$

The maximum over routes in Equation 4.2 is what allows a legally sufficient alternative to count as success. Any-gold-child recall records whether at least one child belonging to any acceptable route appears; parent recall records whether a child of an acceptable legal parent appears. MRR is

$$\text{MRR} @k = \frac{1}{N} \sum_{j=1}^N \frac{1}{r_j}, \quad (4.3)$$

In Equation 4.3, r_j is the rank of the first acceptable child, contributing zero when none occurs within k . nDCG additionally rewards acceptable children appearing earlier, crediting graded rather than binary relevance and discounting by position [52].

Each experiment stores these measures at cutoffs $k \in \{1, 3, 5, 10, 20\}$, separately for the development partition, the test partition, and all questions, together with the denominator n in each view.

The experiment registry now uses the machine-readable keys written by the runs: any-gold-child recall, mean evidence recall, parent recall, MRR, nDCG, and the denominator n , at cutoffs 1, 3, 5, 10, and 20.

4.8.2 Generation Measures

Generation is evaluated through three separate families, which are never combined into one score.

Structural measures record the valid-schema rate, the allowed-source-only rate, the quote-verification rate, and canonical citation accuracy. Answer-quality measures record claim support, completeness, and exact legal-answer accuracy. Selective measures record coverage, risk at coverage, and appropriate-abstention rate. Operational measures — latency, input and output tokens, and estimated cost — are recorded alongside, since an

escalation policy that improves accuracy at unbounded cost has not solved the stated problem.

Surface-overlap metrics are reported only as continuity with the historical baseline and are never presented as evidence of correctness. Recall-oriented n-gram overlap [55] correlates only weakly with human faithfulness judgements, while entailment-based measures correlate considerably better [56]. Citation precision and recall are reported separately from answer correctness, following the benchmark practice established for citation-bearing generation [45].

4.8.3 Statistical Treatment

Proportions are reported with score-based confidence intervals rather than normal-approximation intervals, which are unreliable at the per-condition sample sizes here and at proportions near zero or one [64]. Comparisons between systems on the same questions are treated as paired, and the count of questions each system uniquely passes is reported alongside the difference in aggregate rates, because two systems can post similar totals while failing on disjoint sets. Subgroup results with small denominators are reported with their denominators visible and are not used to support general claims.

4.9 Threats to Validity

Table 4.8 states the principal threats to the validity of this study and the control applied to each.

4.10 Reproducibility and Audit Trail

Experiment directories are immutable by default. Each manifest records input and output digests, parameters, software and framework versions, model identifiers and revisions, hardware, timestamps, and a status field. Dense vectors and cross-encoder scores are cached under their input digests so that a repeated run is verifiably the same computation rather than a similar one. Determinism is achieved through explicit tie-breaking and hash-derived ordering rather than through seeded random number generators, so that results reproduce across platforms and library versions.

The test strategy requires tests before data or results may be promoted, covering corpus construction, retrieval behaviour, evaluation semantics, and safety properties. The corpus audit records twelve parser and integrity tests passing at promotion, together with a byte-identical independent rebuild.

The corpus builders, the retrieval and generation drivers, the evaluator, the verification layer and the test suite are published in full at <https://github.com/MoSaleh47/>

Table 4.8 – Principal threats to validity and the control applied to each.

Threat	Control
Benchmark provenance	Questions and reference answers were machine-generated; a stratified sample is audited by a panel of practitioners and agreement is reported [58].
Evidence leakage across splits	Partitioning by legal parent, with a programmatic assertion that no group spans both partitions.
Contamination of public benchmarks	A benchmark built over one specific instrument rather than reused from a public test set [60].
Reranker domain shift	The training distribution is stated, and reranking gains are interpreted as obtained despite that shift [33].
Threshold tuning on test data	All selection and calibration performed on development; fusion uses an untuned constant.
Metric conflation	Retrieval, structural, semantic, selective, and operational measures reported separately.
Corpus defects mistaken for model limits	Corpus repaired and validated before any retrieval error is attributed to a model.
Single-instrument scope	Results are stated for one regulation; no claim of generalisation to other instruments is made.
Author involvement in review	Panel ratings reported with and without the author’s own; contested records adjudicated openly and reported separately.

Where-RAG-Breaks---Citation-Grounded-Question-Answering-and-Escalation-over-EU-Avi
 with a README describing the pipeline and the order in which the scripts are run;
 Appendix D reproduces the excerpts the argument depends on most directly.

Decisions are append-only and each reversal references the decision it supersedes. Rejected ablations remain documented. Every run computed before the freeze carries an explicit review-pending status in its manifest and is kept apart from the runs computed on frozen labels; the historical baseline is likewise preserved separately from the repaired pipeline, so that none of the three is silently conflated with another.

4.11 Ethical and Professional Considerations

Three commitments constrain the method. Transmission of corpus and question text to a hosted embedding service was treated as requiring explicit authorisation, was refused until that authorisation was given, and is recorded with the token counts actually processed (decisions D012 and D013). Reranking runs locally precisely so that no project text reaches a hosted inference endpoint at that stage. And the AI-assisted curation of the benchmark is disclosed in the reporting configuration and in Appendix B rather than presented as human-authored gold data.

The system is a decision-support research prototype. It is not a compliance decision-

maker, and any output it produces requires verification by a qualified person before it is relied upon. This constraint is stated in the user-facing output and is tested as a safety property, not left as a disclaimer in the documentation.

Chapter 5

Results and Analysis

Validity status. All retrieval, baseline and benchmark figures in this chapter are computed against the *frozen, independently reviewed* benchmark of 98 records described in Section 5.2. Generation results (Section 5.5) are reported on **both partitions**: the held-out run was made once, after every component it evaluates had been frozen, and nothing was revised in response to it. Escalation calibration (Section 5.6) remains **development-only**, because the one component it would fit — the retrieval-uncertainty threshold — is deliberately left uncalibrated. Every figure is reproduced from a stored experiment manifest, metrics file or validation record; where a quantity exists only in a written report, the text says so.

5.1 Overview

This chapter reports what the system does and where it fails. It proceeds from the data outwards: the corpus and the benchmark (Section 5.2), retrieval (Section 5.3), the historical three-system baseline re-scored under a corrected evaluator (Section 5.4), evidence-constrained generation and its verification (Section 5.5), and the calibration of the escalation policy (Section 5.6). Section 5.7 isolates two properties of the benchmark that no internal metric of the pipeline can detect, and Section 5.8 states the threats to validity.

Table 5.1 collects the chapter’s principal results with a pointer to where each is established.

Three results are stated in advance because they shape what follows; none of them is a performance number.

1. Four aviation professionals shown the same benchmark records agreed on the overall verdict for fewer than half of them (Section 5.2.2). A single gold label was never as certain as a single number implied.
2. The system’s two failure modes — selecting the wrong provision, and misreporting what the selected provision says — did not co-occur even once on the development set (Section 5.6). They are independent, and detecting one tells the system nothing about the other.

Table 5.1 – Principal results of this chapter, with the section or table in which each is reported. Held-out figures come from a single run made after every component they evaluate had been frozen.

Quantity	Value	Reported in
<i>Corpus and benchmark</i>		
Legal parents / retrieval children	109 / 1,535	§5.2
Frozen benchmark after independent review	98 records	Table 5.4
development / held-out partition	42 / 56	Table 5.4
legal parents shared between partitions	0	Table 5.4
Reviewer agreement, Fleiss’ κ , three categories	0.250	Table 5.3
<i>Retrieval, held-out ($n = 56$)</i>		
Hybrid with cross-encoder, recall @1 / @5 / @20	71.4 / 89.3 / 96.4%	Table 5.6
Depth-20 recall gained by reranking	none	§5.3
Mean evidence recall at depth 5	81.1%	§5.3
<i>Historical baselines re-scored ($n = 98$; LoRA at 73)</i>		
Best system (RAG, $k = 5$), parent-level citation	0.625	Table 5.8
The same system, exact citation	0.443	Table 5.8
The same system on the 25 records the draft excluded	0.360	Table 5.9
Few-shot $k = 10$, original scorer then exact	0.286 / 0.159	Table 5.8
<i>Evidence-constrained generation, 140 answers</i>		
Quote verification passed, dev / held-out	95.2 / 87.5%	Table 5.10
Answers containing an elided quote, dev / held-out	0.0 / 7.1%	Table 5.10
Evidence selection correct, dev / held-out	78.6 / 83.9%	Table 5.10
Schema contract satisfied, dev / held-out	100 / 98.2%	Table 5.10
Abstentions	0 of 140	§5.5.6
Enriched context: evidence selection, input tokens	81.0% / +79%	Table 5.11
<i>Escalation</i>		
Development error base rate	21.4% (9 of 42)	§5.6
Best candidate rule, precision [95% CI]	0.42 [0.23, 0.64]	Table 5.12
Questions required to demonstrate it	≈ 178	Table 5.13
Answers failing in both modes	0	Figure 5.2

3. Given a schema field, an explicit instruction, and no cost for using it, the generator declined to abstain on every one of 140 opportunities across both partitions, including on answers that were demonstrably ungrounded (Section 5.5.6). Abstention cannot be delegated to the generator’s own judgement.

5.2 The Corpus and the Frozen Benchmark

5.2.1 Structural corpus

The parent–child corpus contains 109 legal parents and 1,535 retrieval children, drawn from Regulation (EU) No 376/2014 together with its implementing and delegated instruments and the associated Guidance Material [4], [5]. The build is deterministic: rebuilding from the same source export reproduces the corpus byte for byte, and the structural validation records zero issues. Short units are preserved rather than merged, because a two-line definition is a meaningful legal object and blind minimum-length merging would reintroduce the boundary confounding that the corpus exists to remove.

5.2.2 The multi-reviewer validity gate

The benchmark was AI-assisted in construction and AI-audited for evidence linkage. That is not a validity claim, and no amount of internal consistency makes it one. A bounded independent review was therefore run: 33 records — every audit exception plus a deterministic stratified sample of unchanged records — were put to practising aviation professionals through a bilingual form, under an aggregation rule fixed before any response arrived.

Four reviewers responded. Table 5.2 describes them.

Table 5.2 – Independent review panel. Reviewers are pseudonymised; the mapping to identities is held separately and is not part of this work.

Reviewer	Records rated	Experience	Familiarity with 376/2014
R1	22	over 10 years	knows the regulation
R2	12	over 10 years	works with it directly
R3	29	2–5 years	works with it directly
R4	33	over 10 years	works with it directly

All 33 records received at least one rating and 31 received at least two. Agreement on the overall verdict is reported in Table 5.3.

Both κ values fall in the “fair” band of the Landis and Koch scale [63], computed with the variable-rater generalisation of Fleiss’ statistic [62], [66] because coverage per record is uneven. The second value collapses the three verdict categories into a binary

Table 5.3 – Inter-rater agreement on the overall verdict, computed over the 31 records with at least two raters.

Quantity	Value
Records with ≥ 2 raters	31
Exact agreement on the verdict	13/31 = 41.9%
Fleiss' κ , three categories	0.250
Fleiss' κ , accept versus needs-change	0.332

accept-versus-change distinction. That collapse was decided *after* seeing the data and is therefore reported as a post-hoc transformation, not as the headline.

The 41.9% figure could be read as a failure of the review. That is not the reading the data supports. Reviewer severity differed substantially — individual acceptance rates ranged from 33% to 69%, and two reviewers never once voted to exclude a record. The disagreement is concentrated on precisely the records that the internal audit had already flagged as difficult. The reading that follows is the less comfortable one: **if four professionals who work with this regulation cannot agree whether an answer is supported by a provision, then a single gold label for those records was never as certain as a single accuracy figure suggested.** This is the empirical foundation for the abstention argument developed in Section 5.6, and it is an argument this work can make only because the review was run.

5.2.3 What the review changed

Fourteen records were cleared by the panel under the pre-registered rule. The remaining 19 were adjudicated by the author, each with a recorded one-line justification, and reported separately from the panel-cleared set. The freeze outcome is given in Table 5.4.

Table 5.4 – Benchmark freeze outcome.

Quantity	Value
Records entering review	33
accepted	19
corrected	12
excluded	2
Benchmark before freeze	99
Benchmark after freeze	98
Development / held-out partition	42 / 56
Distinct legal parents	60
Parents appearing in both partitions	0

Two records were excluded. QA_0003 was already marked invalid by the audit and the panel confirmed it. QA_0063 — one of the three residual retrieval failures identified

before the review — was accepted by no reviewer at all, splitting two-to-two between “correct” and “exclude”. The prior hypothesis that this record was defective rather than a system failure was therefore confirmed from outside the system.

The two other residuals moved in the opposite direction. QA_0080 and QA_0176 were both *accepted* by the panel, which means the retrieval misses on those records are genuine system failures and cannot be explained away as incomplete gold labels. An earlier internal analysis had argued that QA_0080 had equivalent evidence elsewhere in the annex; the panel disagreed, and the panel’s verdict stands.

One methodological point deserves emphasis because it was nearly missed. The freeze script originally recomputed the parent-grouped partition from the retained records. Because that procedure fills groups greedily toward a target proportion, removing a single record reordered the groups and moved **12 records across the development/held-out boundary**. Every earlier selection decision — the embedding model, the fusion configuration, the reranker — had been made on the previous development set, so records used for selection would have landed in the held-out partition. The partition is therefore *inherited* rather than recomputed: no record changed side, and every earlier selection remains valid. The observed development fraction is consequently $42/98 = 0.429$ rather than the nominal 0.400, and it is reported as observed.

5.3 Retrieval

All figures in this section are computed against the frozen reviewed labels on the inherited partition. Selection decisions were made on development data; held-out figures are reported after the fact and were not used to choose anything.

5.3.1 Dense embedding comparison

Both embedding models index identical children and identical questions. Table 5.5 reports the development criterion on which the selection was made.

Table 5.5 – Development recall for the dense embedding comparison, reviewed labels. Proportion of questions retrieving at least one acceptable evidence child.

Dense model	@5	@10	nDCG@10
text-embedding-3-small	66.7%	78.6%	0.504
text-embedding-3-large	71.4%	76.2%	0.536

The larger model was selected. On reviewed development labels it leads on five of the six criteria examined — recall and nDCG at depth 5, evidence recall at depths 5 and 10, and nDCG at depth 10 — but it *loses* recall at depth 10, 78.6% against 76.2%. Under the pre-review labels it led at both depths.

This reversal is reported rather than smoothed over, and the selection is not re-made. Re-choosing the embedding model after seeing reviewed labels would make the choice depend on the same data that grounds the reported results. The operating point of the deployed configuration is depth 5 followed by reranking, where the larger model still leads. The honest summary is that the two embedding models are closer than the original margin suggested, and that a selection justified on a single cutoff would have been fragile.

5.3.2 Lexical, hybrid and reranked retrieval

Table 5.6 reports held-out performance for all five configurations, and Figure 5.1 shows the same data.

Table 5.6 – Held-out retrieval on the frozen reviewed benchmark ($n = 56$). Proportion of questions retrieving at least one acceptable evidence child.

Method	@1	@3	@5	@10	@20
Dense 3-small	51.8%	60.7%	67.9%	80.4%	91.1%
Dense 3-large	46.4%	64.3%	76.8%	83.9%	89.3%
BM25	53.6%	71.4%	75.0%	85.7%	85.7%
Hybrid RRF	55.4%	78.6%	83.9%	91.1%	96.4%
Hybrid + cross-encoder	71.4%	89.3%	89.3%	94.6%	96.4%

Held-out retrieval results on the frozen, independently reviewed benchmark

Reviewed benchmark | test $n=56$ | parent-grouped split inherited from the candidate partition (D028)

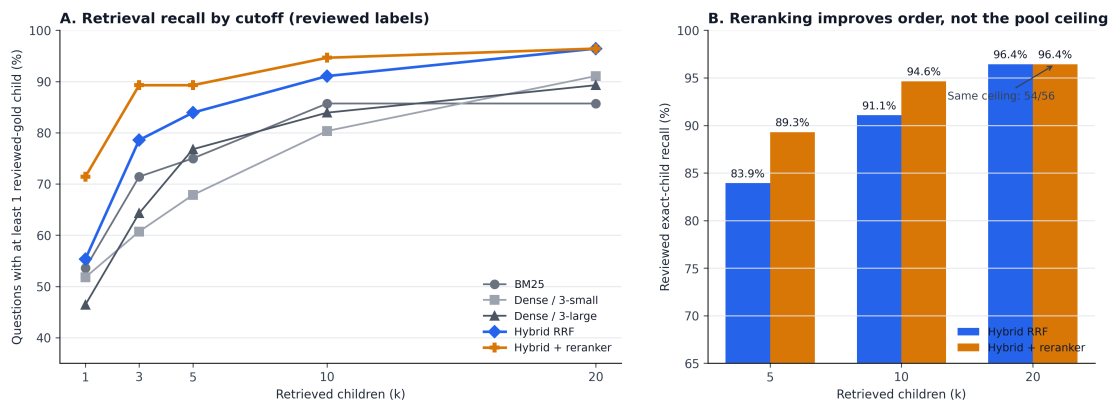


Figure 5.1 – Held-out retrieval on the frozen, independently reviewed benchmark. Left: recall by cutoff for all five configurations. Right: reranking changes the order of the candidate pool without changing its ceiling. Source: stored experiment metrics files.

Three observations follow.

The channels are complementary. BM25 is strong at exactly the places dense retrieval is weak — article numbers, defined terms, structured cross-references — and

dense retrieval recovers paraphrased questions that share no vocabulary with the provision. Equal-weight reciprocal-rank fusion [26] improves on both at every depth. It was chosen because an untuned fusion function has no parameters that could leak information from the held-out partition. This is a deliberate trade of performance for methodological safety, and it should be stated as such: convex combination of normalised scores is known to outperform reciprocal-rank fusion when its weights are tuned [27], and tuning them here would have required fitting on data that also grounds the reported result.

Reranking reorders; it does not enlarge. The cross-encoder raises top-1 recall from 55.4% to 71.4% and depth-5 recall from 83.9% to 89.3%, while depth-20 recall is identical at 96.4%. This is the expected signature of a reranking stage [30]: the candidate pool is fixed by fusion, and the reranker only decides what reaches a small generation budget. Any evidence absent from the top 20 is unrecoverable downstream, which is why the escalation policy in Section 5.6 treats pool exhaustion as a distinct condition from mis-ordering.

Evidence recall trails child recall. Under the mean-evidence-recall measure — the proportion of a record’s required evidence set that is retrieved, rather than whether any of it is — the reranked configuration reaches 81.1% at depth 5 and 85.6% at depth 10 on held-out data. The gap between this and the 89.3% single-child figure is entirely accounted for by records whose gold evidence spans several children of one legal parent.

5.3.3 Effect of the review on measured performance

Because the partition was inherited, the pre-review and post-review runs are directly comparable: same records, same split, one record fewer in the held-out set. Table 5.7 gives the comparison for the deployed configuration.

Table 5.7 – Held-out recall for hybrid retrieval with cross-encoder reranking, before and after independent review.

Labels	@1	@5	@10	@20
AI-assisted, pre-review ($n = 57$)	66.7%	87.7%	93.0%	94.7%
Frozen, reviewed ($n = 56$)	71.4%	89.3%	94.6%	96.4%

Every system improved, by one to five points. This direction matters for the interpretation of the review: the corrections mostly repaired labels that had been marking correct retrievals as misses, rather than exposing failures the earlier labels had concealed. The review could have been presented as having uncovered hidden weakness. It did not, and the direction is reported as observed.

5.3.4 Rejected alternatives

Two simpler interventions were tested on development data and rejected. Parent-mass boosting, which adds accumulated scores from sibling children of the same legal parent, consistently reduced recall: long or repetitive guidance sections accumulate score without any guarantee that the specific provision is relevant. Unequal fusion weights likewise failed to improve the top-20 candidate objective; a BM25-heavy setting recovered one development question at depth 5 while reducing deeper recall.

Both are retained in the record as negative ablations. They establish that the gain from the cross-encoder comes from direct query–passage interaction rather than from hierarchy awareness in general, and that structural knowledge of the corpus is not automatically useful just because the corpus is structured.

5.4 The Three-System Baseline, Re-scored

The earlier comparison of few-shot prompting, retrieval-augmented generation and a LoRA-adapted 3B model [14] is a historical baseline rather than a contribution of this work. It is re-scored here because its original evaluation used a citation measure that does not survive inspection.

5.4.1 What the original measure counted

The original scorer returned a full match whenever the gold citation string occurred anywhere in the prediction, and half credit for naming the correct article number alone; a record counted as correct at half credit or above, combined with a lexical content threshold. Both rules are unsound. Substring containment means a gold citation of “Article 4” matches inside a prediction that says “Article 4(6)”, and “Section 1” matches inside “section 1.2”. More seriously, half credit for the article number passes the correctness threshold, so a prediction citing Article 16(1) where the provision is Article 16(6) was recorded as correct. In citation-grounded legal question answering that is not a near miss; it is the wrong provision, and it is the specific failure the system exists to prevent.

The replacement resolves citations against the corpus rather than comparing strings. Two quantities are reported. *Parent-level* correctness holds when the prediction names a citation that resolves, against the parent corpus, to the provision the gold citation denotes; only identifiers that exist in the corpus can be credited, so an invented reference resolves to nothing. *Exact* correctness additionally requires every atomic unit of the gold citation, including the sub-paragraph, to be named. Where a gold citation does not state which regulation it means, every regulation containing that provision counts

as a candidate: the benchmark’s own label does not make the distinction, and a system is not penalised for reproducing an ambiguity it inherited.

5.4.2 Scope

Of the 98 frozen records, 25 carry a question the stored predictions never saw — 8 reworded during the independent review and 17 reworded earlier by the AI audit. Scoring a stored answer against a question that was never put to the model measures nothing. An earlier draft of this chapter therefore reported the comparison over the 73 records where a stored prediction was still a valid answer, and warned that the exclusion was not neutral: the audit rewrote the questions it judged vague or compound, so the surviving 73 were plausibly the better-posed subset.

That warning is now a measurement rather than a caveat. The few-shot and retrieval-augmented predictions were regenerated against the frozen wording over all 98 records, with the historical system prompts reproduced verbatim, the same exemplar-sampling procedure at seed 42, temperature 0, the same k values and the same legacy flat chunk index. Two things differ from the original run and are recorded rather than hidden: the model is pinned to `gpt-4o-mini-2024-07-18` instead of the floating `gpt-4o-mini` alias, so the comparison cannot change model underneath itself; and retrieval uses NumPy cosine similarity over L2-normalised vectors in place of `faiss.IndexFlatIP`, which computes the same quantity with ties broken by chunk order (decision D049). The comparison below therefore covers **all 98 records**, of which 88 have a gold citation that resolves to a provision at all.

The LoRA system is the exception and is reported at $n = 73$. Re-running it needs the base model weights and a local GPU rather than an API call, and it was not re-run before submission; its stored predictions still answer the pre-freeze wording, so the scorer drops the same 25 records for it. Every table below carries its denominator, and the asymmetry is stated wherever the systems are compared.

5.4.3 Results

The ranking is unchanged. Retrieval-augmented generation beats few-shot prompting, which beats the LoRA-adapted model, under every measure. The conclusion of the earlier comparison survives both the corrected evaluator and the extension to the full benchmark, and it is stated first.

The re-run reproduces the stored predictions where it should. Table 5.9 separates the two things that changed. Holding the record set at 73 and regenerating the predictions leaves RAG essentially where it was: $k = 1$ is identical on every measure, and $k = 3$ and $k = 5$ differ by at most one record. Few-shot moves further, by up to two records, which is expected rather than troubling — its exemplars are drawn at

Table 5.8 – Historical baselines re-scored on the frozen benchmark. Few-shot and RAG were regenerated against the frozen question wording and cover all 98 records, with citation measures over the 88 whose gold citation resolves to a provision. LoRA retains its stored predictions and is therefore scored on the 73 records whose wording did not change, 63 of them resolvable. The loose measure is the original scorer, reproduced verbatim and reported for continuity only.

System	n	Loose (original)	Parent-level	Exact	ROUGE-L
Few-shot, $k = 1$	98	0.235	0.318	0.193	0.246
Few-shot, $k = 5$	98	0.286	0.330	0.159	0.239
Few-shot, $k = 10$	98	0.286	0.318	0.159	0.246
LoRA 3B	73	0.178	0.238	0.111	0.277
RAG, $k = 1$	98	0.429	0.409	0.330	0.338
RAG, $k = 3$	98	0.592	0.568	0.420	0.410
RAG, $k = 5$	98	0.633	0.625	0.443	0.432

Table 5.9 – Where the change from the earlier draft’s figures comes from, at parent level. *Stored*, 73 is the draft’s comparison. *Regenerated*, 73 holds the record set fixed and changes only the predictions, so it measures run-to-run variation. *All 98* holds the provenance of the predictions fixed and adds the 25 records the draft excluded. Denominators are 63 and 88 resolvable records.

System	Same 73 records		
	Stored	Regenerated	All 98
Few-shot, $k = 1$	0.270	0.238	0.318
Few-shot, $k = 5$	0.254	0.302	0.330
Few-shot, $k = 10$	0.302	0.270	0.318
RAG, $k = 1$	0.492	0.492	0.409
RAG, $k = 3$	0.667	0.683	0.568
RAG, $k = 5$	0.730	0.730	0.625

random and the draw sequence changes when the record set does. The middle column is a run-to-run control, and it is tight for the system the argument rests on.

The excluded records were harder for RAG, and the draft’s figures were inflated by their absence. On the 25 records the earlier draft dropped, RAG at $k = 5$ reaches 0.360 [0.202, 0.555] at parent level against 0.730 [0.610, 0.824] on the 73 it kept. The intervals do not overlap. Adding those records costs the best system ten points at parent level and eight at exact. The scope note in the draft predicted exactly this, and the prediction is now confirmed rather than hedged.

They were not harder for few-shot, which points at retrieval as the mechanism. On the same 25 records, few-shot at $k = 10$ moves the other way: exact-citation 0.360 [0.202, 0.555] against 0.079 [0.034, 0.173] on the 73. The questions the audit rewrote for vagueness are ones a retriever handles worse and a model answering from parametric knowledge handles better, which is what one would expect if the difference is mediated by retrieval — few-shot does not retrieve at all. On that subset the ordering of the two systems reverses, though 25 records cannot establish that the reversal is real, and it is reported as a direction rather than a result.

The loose measure still overstates most severely where the dominant failure is a plausible-looking wrong citation. Few-shot at $k = 10$ falls from 0.286 under the original scorer to 0.159 exact, because its characteristic error is to name a real article with the wrong sub-paragraph, which is what the half-credit rule rewarded. Fourteen of its records counted correct under the original scorer cite the wrong sub-paragraph of the right article: Article 18(5) where the provision is Article 18(2), Article 16(1) where it is Article 16(6).

The most informative quantity is the gap within the RAG column. At $k = 5$, retrieval-augmented generation reaches 0.625 [0.521, 0.719] at parent level but 0.443 [0.344, 0.547] exact. Those 18 points are sixteen answers that retrieved the correct article, read fluently, and cited the wrong paragraph of it. The gap was 22 points on the draft’s 73 records and it is 18 on the full benchmark: the correction narrows it without touching the finding. No metric based on substring citation matching can see that gap, and no lexical content measure detects it either. It is the empirical motivation for the verification layer.

ROUGE-L [55] is reported because the original evaluation used it and continuity matters. It is a lexical proxy. It is not evidence of semantic support, and no threshold on it is described in this work as correctness.

5.5 Evidence-Constrained Generation

Generation was run on the development partition ($n = 42$) and then once on the held-out partition ($n = 56$), using `gpt-4o-mini-2024-07-18` at temperature 0. The

held-out run was made only after the prompt, the verification rule, the retrieval stack and the benchmark were all fixed, and no component was revised after seeing its result. Both partitions are reported together throughout; where they disagree, they are shown disagreeing. The generator is held fixed at the model used for the historical few-shot and RAG baselines, so that the comparison between this pipeline and those baselines isolates the scaffolding — retrieval quality, the evidence constraint and the output contract — rather than model capability. A more capable generator would also suppress the failures this chapter exists to study.

5.5.1 A schema-conformant answer is not a contract-valid answer

Across 140 generated answers there were zero parse failures and zero refusals. The request uses the API’s strict structured-output mode [67], derived from the project’s own schema, and every response is then re-validated locally against that schema. On the development partition nothing failed. On the held-out partition, one answer did, and that case bears directly on the argument of this chapter.

For QA_0017 the model attached a claim to a source identifier it was never given:

```
supplied  EU_376_2014_ART_5__C0001
emitted   EU_376_2014_ART_5__C001
```

The reference is fabricated, and it differs from a real one by a single omitted character.

The API’s strict mode had already accepted this response, and correctly so. Strict structured output guarantees the *shape* of a reply: that the field exists and contains a string. It cannot guarantee that the string is one the application supplied, because that is not a property of the schema — it is a property of the request. The distinction is invisible in normal operation and becomes visible exactly when a model produces a plausible near-miss, which is also precisely the case a human reviewer skimming the answer would not catch. The deterministic allowed-source check caught it immediately.

Two consequences follow. Structural conformance is otherwise not a confound: 139 of 140 answers were well formed, so the failures reported below are failures of grounding rather than of format. And provider-side structured output does not remove the need for application-side validation of the values it carries; the architecture keeps local re-validation after every generation call regardless of what the provider guarantees.

5.5.2 What counts as a verbatim quote

The contract requires every claim to carry a source identifier and a verbatim supporting quote, and forbids the model from writing display citations at all — those are resolved by the application from the identifier. Verification of the quote is deterministic, but the

definition of “verbatim” is a design decision that moves the reported grounding rate by twenty-one points, so it is stated explicitly.

Of 132 quote spans in the first run, 90 were byte-exact. **Zero** were repaired by normalising unicode or typographic quotation marks — so typography is not what breaks these quotes. Thirty-one contained exactly the same words in the same order and differed only in punctuation, almost all being the model ending a quote early and closing it with a full stop where the source continued with a comma. Eleven failed genuinely.

A quote is therefore accepted when its word sequence appears *contiguously* in the cited source after punctuation and typographic normalisation. Punctuation is ignored because the corpus is a PDF extraction that carries its own artefacts — **safetyrelated** appears unhyphenated in the source text — so requiring byte-identical punctuation partly measures the extractor. Contiguity is *not* relaxed, and it is the condition that does the discriminating: it is what rejects a quote assembled from fragments. The byte-level result is reported alongside throughout as a second figure.

5.5.3 One prompt rule, measured

The first prompt asked for a “verbatim supporting quote” without saying what that excluded. It produced elided quotes — quotes with material omitted behind an ellipsis — in 7 of 42 answers. In every case the omission removed part of a legal test. On QA_0012, the model quoted a provision on unmanned aircraft as “...not required... unless the occurrence... resulted in a fatal or serious injury to a person”, deleting both the cross-reference identifying which aircraft the rule covers and an entire alternative limb of the exception. The resulting answer is fluent, cites the correct article, and states the exception more narrowly than the law does.

A second prompt version changes exactly one rule: quote one unbroken run of consecutive words, never use an ellipsis, never join two passages, and if the required span is long then quote less or provide several separate supports. Nothing else differs. Table 5.10 reports the effect.

Nine development records were repaired for the cost of one sentence of prompt. The claim count fell from 124 to 114 while the number of quote spans held steady at 132 and 133, which suggests the constraint suppresses claims the model could not ground rather than degrading the answers.

The rule did not transfer as cleanly as development suggested. On development it eliminated elision entirely — 7 answers to 0. On held-out questions it did not: 4 of 56 answers still contain an elided quote. The development zero was optimistic, and the statement the data supports is that the rule substantially reduced the failure without removing it. Grounding fell from 95.2% to 87.5%; the Wilson

Table 5.10 – Effect of a single prompt rule forbidding elided quotes, and how it transfers to unseen questions.

Quantity	Development ($n = 42$)		Held-out ($n = 56$)
	v1	v2	v2
Quote verification passed	76.2%	95.2%	87.5%
under the byte-level rule	54.8%	73.8%	64.3%
Answers containing an elided quote	16.7%	0.0%	7.1%
Evidence selection correct	81.0%	78.6%	83.9%
Schema contract satisfied	100%	100%	98.2%
Model abstained	0	0	0

intervals overlap ($[0.842, 0.987]$ against $[0.764, 0.938]$), so that particular drop is not distinguishable from sampling variation at this size, but the reappearance of elisions is a qualitative recurrence rather than a question of interval width.

The simplest intervention that addresses the failure was tried first and measured rather than assumed, and measuring it on unseen data is what showed that a fix validated on development can look complete and not be. That is an argument for keeping the verifier in place permanently rather than trusting the instruction that was meant to make it unnecessary.

5.5.4 The failure the fix displaced

The second prompt removed every *marked* elision. It introduced one *unmarked* one.

On QA_0294, instructed not to write an ellipsis, the model did not quote a shorter continuous span. It joined two fragments with no marker at all, producing “who have the role to verify compliance with applicable design data...” from a source reading “...who are directly involved in the production of aeronautical items, have the role to verify compliance with applicable design data...”. The omitted clause is the one that limits the provision to persons directly involved in production.

The consequence is methodological. A syntactic check for ellipsis characters would have passed this answer; only verifying that the word sequence actually appears contiguously in the source rejects it. **A prompt constraint displaced the failure mode into a less visible form rather than removing it.** This is the argument for verification over instruction, and it is the argument this project makes generally: a rule stated to a generator is a request, whereas a rule checked against the evidence is a guarantee.

The one other surviving failure is plainer. On QA_0047 the model asserted that “Member States and EASA are not allowed to record personal details in their databases”, attributing it to a source that says only that organisations must keep occurrence details confidential. The same structural check rejects it for the same reason.

5.5.5 The enriched context variant

Chapter 4 specifies two context variants and everything reported so far uses the compact one: the five reranked children under a 1,200-word budget. The enriched variant keeps the same five anchors and adds their sibling units at radius one. It is evaluated here on the development partition only. A second post-freeze run on the held-out partition would retire the single-run discipline established at the start of this section, and an ablation is not worth that (decision D050).

Table 5.11 – Compact against enriched context on the development partition ($n = 42$). Prompt, schema, model, temperature and question set are identical; only the context construction differs.

Quantity	Compact	Enriched
Quote verification passed	40 (95.2%)	40 (95.2%)
under the byte-level rule	31 (73.8%)	32 (76.2%)
Answers containing an elided quote	0	1
Evidence selection correct	33 (78.6%)	34 (81.0%)
Schema contract satisfied	42 (100%)	42 (100%)
Model abstained	0	0
Claims made	114	126
Quote spans	133	140
Distinct sources cited per answer (mean)	2.60	2.98
Prompt tokens	70,089	125,205
Completion tokens	13,747	14,673

The enriched context costs 79% more input tokens and returns one record on evidence selection — 34 of 42 against 33, or 0.810 [0.667, 0.900] against 0.786 [0.641, 0.883] — and one on byte-exact quoting. It also reintroduces an elided quote where the compact run had none. At this sample size those movements are each a single answer, and the Wilson intervals overlap almost entirely; the only difference measured with any confidence is the cost.

One pattern is worth naming because it is not a matter of interval width. The enriched run makes more claims (126 against 114), cites more distinct sources per answer (2.98 against 2.60) and produces more quote spans (140 against 133). More evidence in the window produced more assertions rather than better-grounded ones: the grounding rate is identical at 95.2% and the elided quote reappeared. This is consistent with the reason the budget exists at all — evidence is used unreliably as the window grows [29] — but it is one development run and is offered as an observation, not a finding.

The variant is therefore specified, executed and reported, and it is not adopted. The compact context remains the configuration on which the held-out result stands.

5.5.6 The generator never abstains

Across all three runs — 140 answers, on both partitions — the model abstained **zero** times.

This is not an oversight in the design. The output schema provides four abstention reasons; the prompt instructs the model to use them when the evidence is insufficient, ambiguous, conflicting or outside the corpus; strict structured output guarantees that returning one is well-formed and costs nothing. It did not do so on the record where it fabricated a prohibition, nor on the record where it spliced away a limiting clause, nor on the record where it invented a source identifier one character away from a real one, nor on any of the eighteen records across both partitions where it attached its claims to the wrong provision.

The literature on selective prediction assumes a model can be induced to express when it should not answer, whether through calibration of its own confidence [49], [50] or through selective answering mechanisms [47], [48]. On this task, with this model, the explicit and freely available channel for doing so was never used. **Abstention therefore cannot be delegated to the generator’s judgement; it must be driven by external verification.** That converts the abstention component of the architecture from a design preference into a measured necessity.

5.6 Escalation Calibration

The escalation policy asks which *runtime-observable* condition justifies declining to answer. A signal qualifies only if it can be computed when the question arrives, without gold labels: retrieval scores, the verifier’s output and the shape of the generated answer all qualify; whether the answer selected the correct provision does not, since that is the outcome being predicted.

5.6.1 The sample cannot support a threshold

On the development partition, 9 of 42 answers selected the wrong provision — a base error rate of 21.4%. Table 5.12 reports the candidate rules with Wilson intervals [64] on precision.

Two rules have a lower confidence bound fractionally above the 0.214 base rate, both at 0.23. That is a two-point margin estimated from nine error events, and one of the two rests on five escalations in total. Neither is a basis for a threshold.

No escalation threshold is fitted in this work. A threshold chosen from nine events would not be distinguishable from noise at this sample size. The policy is specified; its retrieval-uncertainty thresholds are left explicitly open.

Table 5.12 – Candidate escalation rules on the development partition ($n = 42$, 9 error events). Precision intervals are Wilson 95%.

Escalate when	Escalated	Precision	Recall
Reranker moved top-1 by ≥ 2 ranks	19	0.42 [0.23, 0.64]	0.89
Only one claim was made	5	0.60 [0.23, 0.88]	0.33
Reranker moved top-1 by ≥ 5 ranks	7	0.43 [0.16, 0.75]	0.33
Rerank margin ≤ 0.5	14	0.36 [0.16, 0.61]	0.56
Rerank margin ≤ 1.0	18	0.33 [0.16, 0.56]	0.67
Model cited ≤ 2 distinct sources	21	0.29 [0.14, 0.50]	0.67
Quote verification failed	2	0.00 [0.00, 0.66]	0.00
Model abstained	0	—	0.00

5.6.2 How much data would be required

Table 5.13 gives the two-proportion sample size, at $\alpha = 0.05$ and 80% power, needed to distinguish a given escalation precision from the observed base rate.

Table 5.13 – Sample size required to demonstrate a given escalation precision against the observed 21.4% base error rate.

Target precision	Escalated decisions	Questions
0.35	172	≈ 383
0.42 (best observed)	80	$\approx$ 178
0.50	43	≈ 96
0.60	25	≈ 56

Question counts assume the 45% escalation rate observed for the most promising rule. Against a frozen benchmark of 98 records, demonstrating the effect actually observed would require roughly 1.8 times the current benchmark. This is a quantified limitation with a stated remedy, reported in place of a threshold fitted to nine events.

5.6.3 The two failure modes are independent

The result that does hold at this sample size is structural rather than numerical, and it is shown in Figure 5.2.

The two sets do not intersect. Quote verification has precision 0.00 and recall 0.00 for predicting evidence-selection errors, and the two answers it rejected had entirely correct evidence.

The finding is architectural rather than a shortfall in performance. The system fails in two independent ways. *Evidence selection* — attaching a claim to the wrong provision — is not observable at runtime and must be predicted from retrieval signals, which the preceding subsection shows cannot yet be done reliably. *Quote fidelity* — misreporting what the selected provision says — is deterministically checkable at runtime with no

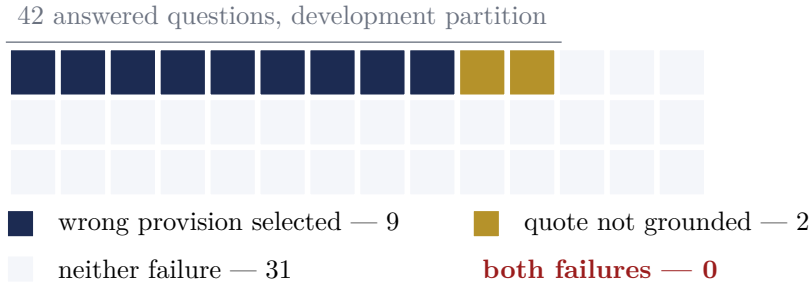


Figure 5.2 – The two failure modes on the development partition, one cell per answered question ($n = 42$). Nine answers selected the wrong provision and two carried a quote that does not appear in the source it cites; eleven answers failed in one way or the other, and none failed in both. Cells are grouped by outcome, so position in the grid carries no meaning.

gold labels at all. A verifier checking only quotes would have passed all nine evidence errors; a check on evidence selection alone would have passed both quote failures. Both layers are required, and the requirement is measured rather than asserted.

5.6.4 The policy as specified

The escalation policy that these results support is stated as follows.

1. Retrieve a broad candidate pool through lexical and dense fusion.
2. Rerank when evidence may be present but poorly ordered.
3. Add adjacent children when a selected provision lacks local continuity.
4. **Block the answer when quote verification fails.** This requires no calibration: a quote either appears contiguously in the cited source or it does not, so the check has no false positives by construction. It fires on 2 of 42 development answers.
5. Request clarification when several legitimate provisions compete.
6. **Escalate on retrieval uncertainty — threshold uncalibrated.** Reranker-versus-fusion disagreement is the most promising observed signal and is named as the candidate, with its threshold left open pending the sample size in Table 5.13.
7. Refer to a human when evidence remains absent, ambiguous or conflicting.

Model self-abstention is excluded as a signal entirely, for the reason given in Section 5.5.6.

Escalation precision and recall are reported together throughout, never separately. A rule escalating 45% of questions to catch 89% of errors may well be correct for aviation regulation, where a wrong citation is expensive and a human referral is cheap — but that is a judgement about the relative cost of the two outcomes, not something the data decides. Wherever an escalation rate appears, the assumed cost ratio is stated with it.

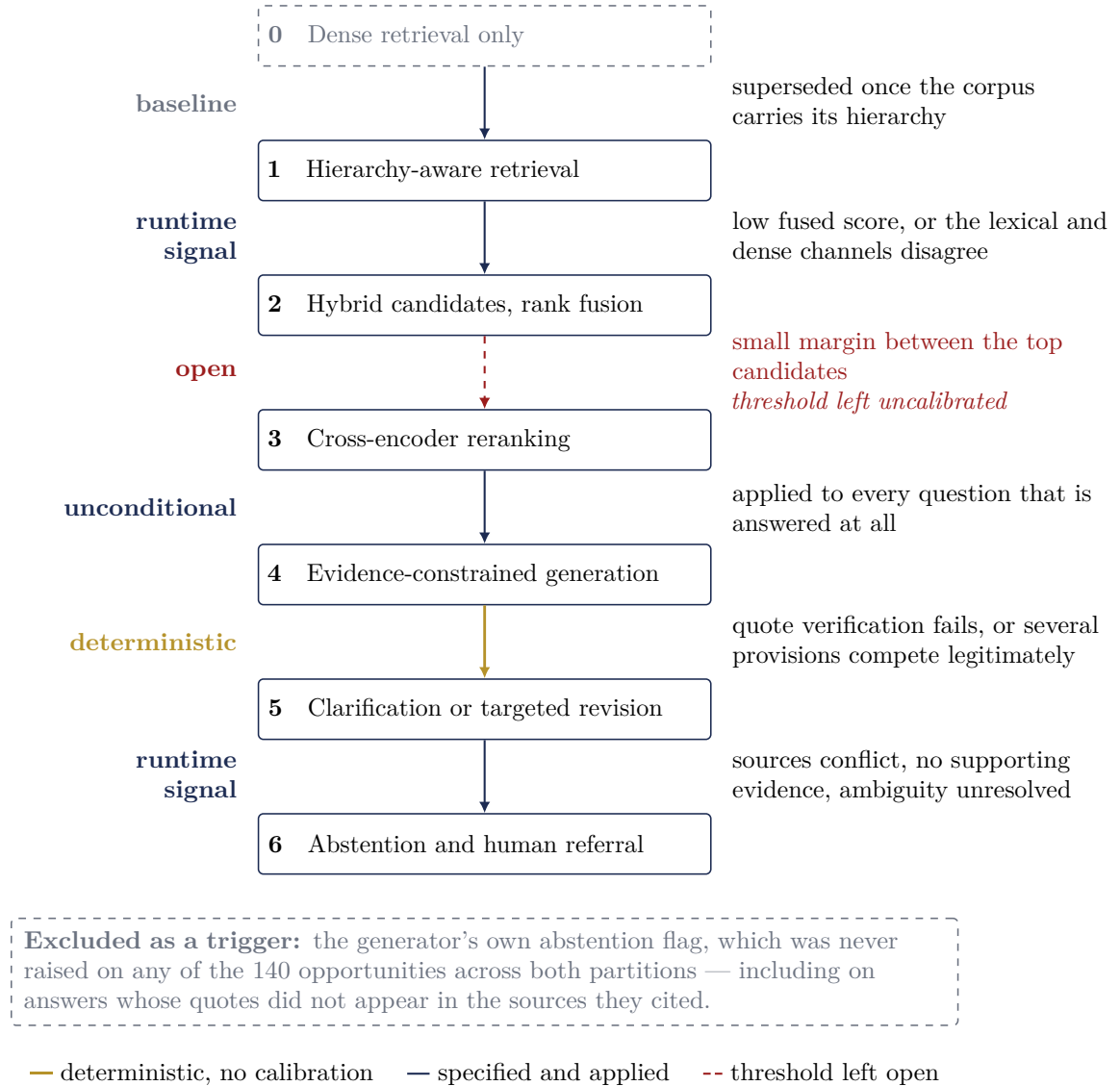


Figure 5.3 – The escalation policy, with the trigger for each transition and what the results of this chapter establish about it. One transition needs no calibration, since a quote either appears contiguously in the cited source or it does not. One is left deliberately open: reranker-versus-fusion disagreement is the most promising observed signal, but nine error events cannot fix its threshold (Table 5.13).

The generator's own abstention flag is excluded as a signal altogether.

5.7 Two Properties No Internal Metric Detects

5.7.1 Citation labels a reader cannot resolve

Six records drew the same reaction from reviewers, in different words each time: “*GM 3.1 = ?*”, “*qu’entends-tu par ‘section 2.2’?*”. A practitioner who works with this regulation daily could not resolve the canonical citation strings.

That observation was then measured rather than left anecdotal. Every gold citation in the frozen benchmark was resolved against the corpus, with the result in Table 5.14.

Table 5.14 – Resolution of the benchmark’s own gold citation labels against the corpus ($n = 98$).

The gold citation. . .	Records
identifies the parent holding its own evidence	48
names an Article while the evidence sits in a Guidance Material section	28
omits which of the four regulations it means	12
cannot be parsed into any provision at all	10

Twenty-two of 98 records carry a citation label that cannot be resolved to a single provision from the string alone — roughly four times what the six free-text comments suggested. A bare “Article 2” exists in all four regulations in the corpus; “GM to Reg. (EU) No 376/2014 [04]” identifies nothing at all.

Every one of these citations resolves correctly *inside* the pipeline, because the pipeline carries the parent identifier alongside the display string. The defect is invisible to every internal metric by construction, and visible immediately to an outside reader holding only what the user is actually shown. This is the clearest available demonstration that internal consistency is not correctness, and it was found by asking practitioners rather than by measuring.

The 28 guidance-material records are a separate and legitimate phenomenon: the citation points to the binding provision while the evidence sits in the guidance discussing it. It is not an error, but it means that “cited correctly” and “retrieved the evidence” are different questions on nearly a third of the benchmark, and results must not silently conflate them.

5.7.2 The benchmark cannot demonstrate an agent’s benefit

Every one of the 98 frozen records has its gold evidence inside a single legal parent. Zero records require evidence from two or more.

This is decisive for the question of agentic, multi-step retrieval. Whatever an agent’s merits in general, a benchmark in which no question requires combining provisions from different parts of the instrument cannot exhibit them: the measurement has no room

to show a difference. Building an agent and reporting a null result on this benchmark would be uninformative, and reporting a positive one would be evidence of something other than the agent’s retrieval behaviour.

A cross-reference evaluation subset is therefore a prerequisite for that line of work, not an optional refinement — a conclusion reached from a property of the data rather than from a position on agent architectures.

5.8 Threats to Validity

Benchmark provenance. The benchmark is synthetic in origin and may reflect the style of the model that generated it [60]. The independent review mitigates but does not eliminate this: 33 of 98 records were reviewed, and the remaining 65 retain AI-assisted evidence linkage.

Agreement is fair, not strong. Fleiss’ κ of 0.250 means the reviewed labels themselves carry disagreement. This is reported as a finding rather than a nuisance, but it bounds how precisely any accuracy figure computed against those labels can be interpreted.

Small samples. The escalation analysis rests on 9 error events on 42 development records, and every precision figure in Section 5.6 is indicative rather than estimated. Generation is reported on both partitions, but 56 held-out records still give wide intervals: the 87.5% grounding rate carries a 95% interval of [0.764, 0.938].

Selection after observation. The second prompt version was chosen after seeing the first version’s failures on development data. That is the legitimate use of a development partition, and the held-out run was that prompt’s first and only exposure to unseen questions. It is reported exactly as observed, including where it is worse than development; no prompt, threshold or model was revised afterwards.

Baseline scope. The few-shot and RAG comparison now covers all 98 records; LoRA remains at 73, since its stored predictions answer the pre-freeze wording and it was not re-run. Comparisons involving LoRA are therefore made on a subset that is measurably easier for retrieval-augmented systems, which flatters RAG against it rather than the reverse. The LoRA system was measured at one variant, on a 3B model, against hosted models of larger and undisclosed size; it is not a controlled comparison of adaptation methods and is not presented as one.

Lexical proxies. ROUGE-L appears only for continuity with the earlier evaluation. Structural verification establishes that a claim points at supplied evidence and reproduces its wording; it does not establish that the evidence supports the claim. Semantic support remains unmeasured in this work [2], [43].

Fusion configuration. Equal-weight reciprocal-rank fusion is known to be sub-optimal relative to tuned convex combination [27]. It was chosen because an untuned

function cannot leak from the held-out partition. The reported retrieval figures are therefore a conservative estimate of what this architecture can achieve.

Chapter 6

Conclusion and Future Work

6.1 Answer to the Research Question

The question this project set out to answer was not whether retrieval-augmented generation can answer questions about Regulation (EU) No 376/2014. It was *where such a system fails, and which condition observable at runtime justifies escalating instead of answering*. The distinction matters because a single accuracy figure conceals the first question entirely and cannot address the second at all.

The answer has three parts, none of which was the answer anticipated at the outset.

6.1.1 Retrieval is not where this system fails

On the frozen benchmark, hybrid retrieval with cross-encoder reranking places acceptable evidence in the top five results for 89.3% of held-out questions and in the top ten for 94.6%. The candidate pool at depth 20 contains acceptable evidence for 96.4%. Retrieval was the natural suspect, and the measurements do not support it: by the time the generator sees its context, the evidence needed to answer is usually present.

What the retrieval work established is narrower than a percentage and bears more directly on the policy. The lexical and dense channels fail on different questions, so fusion addresses a coverage gap rather than tuning an existing configuration; reranking changes the order of a fixed pool without enlarging it, so anything absent at depth 20 is unrecoverable downstream and must be handled as a distinct condition rather than as a bad answer. Those two facts shape the escalation policy more than the recall figures do.

6.1.2 The failure is citation precision, and it is invisible to conventional measures

The historical baseline, re-scored on frozen labels and regenerated over the whole benchmark, locates the failure more precisely. Retrieval-augmented generation identifies the correct legal parent for 62.5% of questions but reproduces the complete citation,

sub-paragraph included, for only 44.3%. Those eighteen points are sixteen answers that retrieved the right article, read fluently, and cited the wrong paragraph of it.

The original evaluation could not see this. Its citation measure awarded credit for substring containment and half credit for naming the article number alone, so a prediction citing Article 16(1) where the provision is Article 16(6) was recorded as correct. Under the corrected evaluator, few-shot prompting falls from 0.286 to 0.159, because naming a real article with the wrong sub-paragraph was precisely its characteristic error, and precisely what the old measure rewarded.

The ranking of the three historical systems is unchanged. The magnitudes are not, and neither is the population they were measured on: an earlier draft reported this comparison over 73 records, because 25 of the 98 had been reworded after the stored predictions were made. Regenerating those predictions against the frozen wording shows the exclusion was not neutral. On the 25 records that had been dropped, retrieval-augmented generation reaches 0.360 at parent level against 0.730 on the 73 that were kept, with non-overlapping intervals, while few-shot moves the other way. The questions an audit rewrites for vagueness are harder for a retriever and easier for a model answering from memory. All three statements are reported together.

6.1.3 The escalation condition, and the honest limit on it

Two conditions observable at runtime were examined. One works and needs no calibration at all; the other cannot yet be calibrated on data of this size.

Quote verification is a usable gate. Whether a quoted span appears contiguously in the source it cites is decidable without gold labels, without model judgement and without a threshold. It has no false positives by construction. On the held-out partition it rejected 7 of 56 answers, and every rejection was a genuine defect: elided quotes that deleted conditions from legal tests, a paraphrase presented as quotation, and a fabricated source identifier one character removed from a real one.

Retrieval-uncertainty escalation cannot be calibrated on data of this size, and no threshold for it is reported. With nine error events on the development partition, every candidate rule’s confidence interval on precision contains or nearly contains the base error rate. The most promising signal — disagreement between rank fusion and the reranker about which result belongs first — reaches 0.42 precision at 0.89 recall, with an interval of [0.23, 0.64] against a base rate of 0.21. That is a direction to investigate rather than a threshold. Demonstrating the observed effect would require roughly 178 questions against a benchmark of 98.

It is reported as a negative result with a stated sample-size requirement rather than as a number fitted to nine events.

6.2 What the Project Established That It Did Not Set Out to Find

Four results emerged from the work rather than from its design, and they are the ones most likely to transfer beyond this corpus.

6.2.1 Verification and instruction are not substitutes

The same negative result appears three times.

Instructed to quote verbatim, the generator elided material. Instructed explicitly not to elide, it stopped marking omissions with an ellipsis and began splicing fragments silently instead — a failure a syntactic check for ellipsis characters would have passed, and which only contiguity matching rejected. The prompt rule that eliminated elision on the development partition did not eliminate it on unseen questions, where it recurred in 4 of 56 answers. And a response the provider’s strict structured-output mode had already accepted carried a source identifier the application had never supplied.

In each case an instruction shifted the failure rather than removing it, and in each case only a check against the evidence detected what remained. **A rule stated to a generator is a request; a rule checked against the evidence is a guarantee.** It is the project’s central practical claim, and it was reached by trying the cheaper option first and measuring the outcome.

6.2.2 Abstention cannot be delegated to the generator

Across 140 generated answers on both partitions, the model abstained zero times. The output schema offered four abstention reasons, the prompt instructed their use, and strict structured output guaranteed that returning one would be well-formed and cost nothing. It abstained on none of the eighteen answers that cited the wrong provision, none of the answers whose quotes were not grounded, and not on the answer that invented an identifier.

Selective-prediction research generally assumes a model can be induced to signal when it should not answer. On this task, with this model, the explicit and freely available channel for doing so went entirely unused. Abstention here is therefore not a behaviour to elicit but a decision to impose from outside, on evidence the system can check.

6.2.3 The two failure modes are independent

On the development partition, nine answers selected the wrong provision and two failed quote verification. *No answer did both.* Quote verification has precision 0.00 and recall

0.00 for predicting evidence-selection errors.

The consequence is architectural. Misreporting what a provision says is detectable at runtime; selecting the wrong provision is not, and must be predicted from retrieval signals. A verifier addressing only the first would have passed all nine errors of the second. Two layers are required, and the requirement is measured rather than assumed.

6.2.4 A benchmark’s labels can be internally valid and externally unreadable

Twenty-two of the 98 frozen records carry a citation label that cannot be resolved to a single provision from the string alone: twelve omit which of four regulations they mean, and ten cannot be parsed into a provision at all. Every one of them resolves correctly inside the pipeline, which carries the parent identifier alongside the display string. The defect is invisible to every internal metric by construction and immediately visible to a practitioner holding only what the user is shown.

It was found because reviewers were asked, not because anything was measured. Six of them wrote variants of “*GM 3.1 = ?*” in a free-text box. The measurement came afterwards and showed the problem to be roughly four times larger than the comments suggested.

6.3 Contributions

1. **A deterministic, hierarchy-aware corpus** of 109 legal parents and 1,535 retrieval children with byte-identical rebuild and zero validation issues, together with explicit lineage from the historical flat chunks.
2. **An independently reviewed benchmark of 98 evidence-linked questions**, split by legal parent with zero leakage, frozen under a pre-registered rule with the review’s disagreement reported rather than resolved away. Fleiss’ κ of 0.250 across four practitioners is itself a finding: a single gold label was never as certain as a single accuracy figure implied.
3. **A citation evaluator that resolves references against the corpus** rather than matching strings, reported at two levels of strictness, with the scorer it replaces retained and its errors pinned by tests so the change is auditable rather than asserted.
4. **A deterministic verification layer** that checks source admissibility and quote contiguity without gold labels or model judgement, and the measurement showing why it must sit alongside, not instead of, a check on evidence selection.

5. **An escalation policy specified against runtime-observable conditions**, with the one component that could be calibrated left explicitly uncalibrated and the required sample size stated.
6. **A methodological record** of 45 append-only decisions and a disclosure log of every AI-assisted artefact with what was verified and what remained outstanding, sufficient for the work to be reconstructed and challenged.

6.4 Limitations

The benchmark is small and synthetic in origin. Ninety-eight records, 33 of which were independently reviewed, is enough to characterise failure modes and not enough to calibrate a policy against them. Its questions were generated by a language model and may reflect that model’s style.

Reviewer agreement is fair, not strong. A κ of 0.250 bounds how precisely any figure computed against these labels can be read. This is reported as a property of the task rather than a defect of the panel, but it is a real limit on interpretation.

Semantic support is unmeasured. Every verification result in this work is structural. That a claim points at supplied evidence and reproduces its wording contiguously does not establish that the evidence supports the claim. The 83.9% held-out evidence-selection figure counts whether the cited source is among the acceptable ones, not whether the sentence built on it is sound.

One model, one language, two configurations, one of them development-only. Generation used a single model at a single temperature. The neighbour-enriched context variant was evaluated on the development partition and not adopted: it costs 79% more input tokens and moves evidence selection and byte-exact quoting by one record each, which 42 questions cannot separate from noise. Retrieval is English-first, and the reranker was chosen for local execution and English passage ranking. Nothing here speaks to multilingual behaviour, which matters for a regulation applied across the Union.

The LoRA arm of the comparison covers 73 of 98 records. Few-shot and RAG were regenerated over the frozen wording and cover all 98; LoRA was not re-run, so its stored predictions are scored on the 73 records whose wording did not change. That subset is measurably easier for retrieval-augmented systems, so any comparison against LoRA flatters RAG rather than the reverse.

Equal-weight fusion is deliberately suboptimal. Convex combination of normalised scores outperforms reciprocal-rank fusion when its weights are tuned. Tuning them would have required fitting on data that also grounds the reported results, so the retrieval figures are a conservative estimate of what the architecture can reach.

This is a research prototype. It does not replace qualified interpretation of aviation law, and nothing in this work should be read as suggesting it could.

6.5 Future Work

Extend the benchmark to the size the calibration requires. The power analysis gives the target directly: roughly 178 questions to demonstrate the escalation precision actually observed, against 98 today. This is the single change that would convert the escalation policy from specified to calibrated, and it is a data-collection task rather than a research one.

Build the cross-reference subset before building an agent. Every record in the frozen benchmark has its gold evidence inside a single legal parent; none requires combining provisions from different parts of the instrument. A benchmark with no multi-hop questions cannot exhibit a multi-hop system’s benefit, so agentic retrieval remains unevaluable here regardless of its merits elsewhere. The prerequisite is a subset of questions whose evidence genuinely spans parents, constructed and reviewed to the same standard as the present set.

Measure semantic support, not only structural grounding. The natural next layer is an entailment check between each claim and the evidence it cites, which would close the gap between “quotes the source correctly” and “is supported by the source”. Given what this project found about delegating judgement to a generator, such a check should be validated against human ratings before being trusted, and its own agreement with those ratings reported.

Repair the citation labels, then re-measure. Twenty-two records carry labels a reader cannot resolve. Rewriting them to name their regulation and provision unambiguously is mechanical, and it would allow the citation-legibility finding to be stated as a repaired defect rather than only as a diagnosis.

Test whether the escalation signals transfer across model capability. The generator was deliberately held fixed at the model used for the historical baselines, so that the comparison isolated scaffolding from capability. Whether a policy calibrated on a weaker generator still holds for a stronger one is an open and practically important question, and running the existing pipeline against a current model would answer it cheaply.

6.6 Closing Remark

The system this work produced answers questions about occurrence reporting with retrieved evidence, checks its own quotations against that evidence, and refuses to answer when the check fails. It is not a system that is right more often than its predecessor by

a large margin. It is a system whose errors can be detected, and detectability is the property this domain requires.

Regulatory questions are asked by people who will act on the answer. For them, a confident wrong citation is worse than a referral to a human, and an answer that cannot be audited is worse than no answer. The finding that bears most directly on deployment is that the generator will not signal when it is in that position — 140 opportunities, zero abstentions — and that the system must therefore be built to determine it independently.

Bibliography

- [1] V. Magesh, F. Surani, M. Dahl, M. Suzgun, C. D. Manning, and D. E. Ho, “Hallucination-free? assessing the reliability of leading AI legal research tools,” *Journal of Empirical Legal Studies*, vol. 22, no. 2, pp. 216–242, 2025. DOI: 10.1111/jels.12413.
- [2] N. F. Liu, T. Zhang, and P. Liang, “Evaluating verifiability in generative search engines,” in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023, pp. 7001–7025. DOI: 10.18653/v1/2023.findings-emnlp.467.
- [3] Y. Gao *et al.*, *Retrieval-augmented generation for large language models: A survey*, 2023. DOI: 10.48550/arXiv.2312.10997. arXiv: 2312.10997 [cs.CL].
- [4] European Parliament and Council of the European Union, *Regulation (EU) no 376/2014 on the reporting, analysis and follow-up of occurrences in civil aviation*, Official Journal of the European Union, OJ L 122, 24.4.2014, p. 18, CELEX: 32014R0376, 2014. [Online]. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:32014R0376>.
- [5] European Union Aviation Safety Agency, *Easy access rules for occurrence reporting (regulation (EU) no 376/2014)*, Revision from December 2022, Cologne, Germany, 2022. [Online]. Available: <https://www.easa.europa.eu/en/document-library/easy-access-rules/online-publications/easy-access-rules-occurrence-reporting>.
- [6] A. Vaswani *et al.*, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186. DOI: 10.18653/v1/N19-1423.
- [8] P. Lewis *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.
- [9] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M.-W. Chang, “REALM: Retrieval-augmented language model pre-training,” in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119, 2020, pp. 3929–3938.

-
- [10] G. Izacard and E. Grave, “Leveraging passage retrieval with generative models for open domain question answering,” in *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, 2021, pp. 874–880. DOI: 10.18653/v1/2021.eacl-main.74.
 - [11] S. Borgeaud *et al.*, “Improving language models by retrieving from trillions of tokens,” in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, PMLR, 2022, pp. 2206–2240.
 - [12] W. Shi *et al.*, “REPLUG: Retrieval-augmented black-box language models,” in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024, pp. 8371–8384. DOI: 10.18653/v1/2024.naacl-long.463.
 - [13] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, “Self-RAG: Learning to retrieve, generate, and critique through self-reflection,” in *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*, 2024. arXiv: 2310.11511.
 - [14] E. J. Hu *et al.*, “LoRA: Low-rank adaptation of large language models,” in *Proceedings of the Tenth International Conference on Learning Representations (ICLR)*, 2022. arXiv: 2106.09685.
 - [15] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient finetuning of quantized LLMs,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 10 088–10 115. arXiv: 2305.14314.
 - [16] O. Ovadia, M. Brief, M. Mishaeli, and O. Elisha, “Fine-tuning or retrieval? comparing knowledge injection in LLMs,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024, pp. 237–250. DOI: 10.18653/v1/2024.emnlp-main.15.
 - [17] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009. DOI: 10.1561/15000000019.
 - [18] T. Formal, B. Piwowarski, and S. Clinchant, “SPLADE: Sparse lexical and expansion model for first stage ranking,” in *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, ACM, 2021, pp. 2288–2292. DOI: 10.1145/3404835.3463098.
 - [19] V. Karpukhin *et al.*, “Dense passage retrieval for open-domain question answering,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 2020, pp. 6769–6781. DOI: 10.18653/v1/2020.emnlp-main.550.

-
- [20] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using siamese BERT-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and IJCNLP*, 2019, pp. 3982–3992. DOI: 10.18653/v1/D19-1410.
 - [21] G. Izacard *et al.*, “Unsupervised dense information retrieval with contrastive learning,” *Transactions on Machine Learning Research*, vol. 2022, 2022. [Online]. Available: <https://openreview.net/forum?id=jKN1pXi7b0>.
 - [22] O. Khattab and M. Zaharia, “ColBERT: Efficient and effective passage search via contextualized late interaction over BERT,” in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, ACM, 2020, pp. 39–48. DOI: 10.1145/3397271.3401075.
 - [23] K. Santhanam, O. Khattab, J. Saad-Falcon, C. Potts, and M. Zaharia, “ColBERTv2: Effective and efficient retrieval via lightweight late interaction,” in *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2022, pp. 3715–3734. DOI: 10.18653/v1/2022.naacl-main.272.
 - [24] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 2318–2335. DOI: 10.18653/v1/2024.findings-acl.137.
 - [25] N. Muennighoff, N. Tazi, L. Magne, and N. Reimers, “MTEB: Massive text embedding benchmark,” in *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, 2023, pp. 2014–2037. DOI: 10.18653/v1/2023.eacl-main.148.
 - [26] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, “Reciprocal rank fusion outperforms condorcet and individual rank learning methods,” in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009, pp. 758–759. DOI: 10.1145/1571941.1572114.
 - [27] S. Bruch, S. Gai, and A. Ingber, “An analysis of fusion functions for hybrid retrieval,” *ACM Transactions on Information Systems*, vol. 42, no. 1, pp. 20:1–20:35, 2024. DOI: 10.1145/3596512.
 - [28] T. Chen *et al.*, “Dense X retrieval: What retrieval granularity should we use?” In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024, pp. 15 159–15 177. DOI: 10.18653/v1/2024.emnlp-main.845.

-
- [29] N. F. Liu *et al.*, “Lost in the middle: How language models use long contexts,” *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024. DOI: 10.1162/tac1_a_00638.
 - [30] R. Nogueira and K. Cho, “Passage re-ranking with BERT,” *arXiv preprint arXiv:1901.04085*, 2019. [Online]. Available: <https://arxiv.org/abs/1901.04085>.
 - [31] R. Nogueira, Z. Jiang, R. Pradeep, and J. Lin, “Document ranking with a pretrained sequence-to-sequence model,” in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 708–718. DOI: 10.18653/v1/2020.findings-emnlp.63.
 - [32] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, “MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5776–5788.
 - [33] T. Nguyen *et al.*, “MS MARCO: A human generated machine reading comprehension dataset,” in *Proceedings of the Workshop on Cognitive Computation: Integrating Neural and Symbolic Approaches (CoCo @ NIPS 2016)*, ser. CEUR Workshop Proceedings, vol. 1773, CEUR-WS.org, 2016. [Online]. Available: https://ceur-ws.org/Vol-1773/CoCoNIPS_2016_paper9.pdf.
 - [34] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras, and I. Androutsopoulos, “LEGAL-BERT: The muppets straight out of law school,” in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 2898–2904. DOI: 10.18653/v1/2020.findings-emnlp.261.
 - [35] L. Zheng, N. Guha, B. R. Anderson, P. Henderson, and D. E. Ho, “When does pretraining help? assessing self-supervised learning for law and the CaseHOLD dataset of 53,000+ legal holdings,” in *Proceedings of the Eighteenth International Conference on Artificial Intelligence and Law*, ACM, 2021, pp. 159–168. DOI: 10.1145/3462757.3466088.
 - [36] I. Chalkidis *et al.*, “LexGLUE: A benchmark dataset for legal language understanding in English,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 4310–4330. DOI: 10.18653/v1/2022.acl-long.297.
 - [37] N. Guha *et al.*, “LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models,” in *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, vol. 36, 2023, pp. 44 123–44 279.

-
- [38] N. Holzenberger, A. Blair-Stanek, and B. Van Durme, “A dataset for statutory reasoning in tax law entailment and question answering,” in *Proceedings of the 2020 Natural Legal Language Processing Workshop*, ser. CEUR Workshop Proceedings, vol. 2645, CEUR-WS.org, 2020, pp. 31–38.
 - [39] A. Tikayat Ray, B. F. Cole, O. J. Pinon Fischer, R. T. White, and D. N. Mavris, “aeroBERT-Classifier: Classification of aerospace requirements using BERT,” *Aerospace*, vol. 10, no. 3, p. 279, 2023. DOI: [10.3390/aerospace10030279](https://doi.org/10.3390/aerospace10030279).
 - [40] A. Miyamoto, M. V. Bendarkar, and D. N. Mavris, “Natural language processing of aviation safety reports to identify inefficient operational patterns,” *Aerospace*, vol. 9, no. 8, p. 450, 2022. DOI: [10.3390/aerospace9080450](https://doi.org/10.3390/aerospace9080450).
 - [41] Z. Ji *et al.*, “Survey of hallucination in natural language generation,” *ACM Computing Surveys*, vol. 55, no. 12, 2023. DOI: [10.1145/3571730](https://doi.org/10.1145/3571730).
 - [42] L. Huang *et al.*, “A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions,” *ACM Transactions on Information Systems*, vol. 43, no. 2, 2025. DOI: [10.1145/3703155](https://doi.org/10.1145/3703155).
 - [43] H. Rashkin *et al.*, “Measuring attribution in natural language generation models,” *Computational Linguistics*, vol. 49, no. 4, pp. 777–840, 2023. DOI: [10.1162/coli_a_00486](https://doi.org/10.1162/coli_a_00486).
 - [44] B. Bohnet *et al.*, “Attributed question answering: Evaluation and modeling for attributed large language models,” *arXiv preprint arXiv:2212.08037*, 2022. DOI: [10.48550/arXiv.2212.08037](https://doi.org/10.48550/arXiv.2212.08037).
 - [45] T. Gao, H. Yen, J. Yu, and D. Chen, “Enabling large language models to generate text with citations,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 6465–6488. DOI: [10.18653/v1/2023.emnlp-main.398](https://doi.org/10.18653/v1/2023.emnlp-main.398).
 - [46] L. Gao *et al.*, “RARR: Researching and revising what language models say, using language models,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 16 477–16 508. DOI: [10.18653/v1/2023.acl-long.910](https://doi.org/10.18653/v1/2023.acl-long.910).
 - [47] Y. Geifman and R. El-Yaniv, “Selective classification for deep neural networks,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4878–4887.
 - [48] A. Kamath, R. Jia, and P. Liang, “Selective question answering under domain shift,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 5684–5696. DOI: [10.18653/v1/2020.acl-main.503](https://doi.org/10.18653/v1/2020.acl-main.503).

-
- [49] Z. Jiang, J. Araki, H. Ding, and G. Neubig, “How can we know when language models know? on the calibration of language models for question answering,” *Transactions of the Association for Computational Linguistics*, vol. 9, pp. 962–977, 2021. DOI: 10.1162/tacl_a_00407.
 - [50] S. Kadavath *et al.*, “Language models (mostly) know what they know,” *arXiv preprint arXiv:2207.05221*, 2022. DOI: 10.48550/arXiv.2207.05221.
 - [51] H. Zhang *et al.*, “R-Tuning: Instructing large language models to say ‘I don’t know’,” in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024, pp. 7113–7139. DOI: 10.18653/v1/2024.naacl-long.394.
 - [52] K. Järvelin and J. Kekäläinen, “Cumulated gain-based evaluation of IR techniques,” *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002. DOI: 10.1145/582415.582418.
 - [53] N. Thakur, N. Reimers, A. Ruckle, A. Srivastava, and I. Gurevych, “BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models,” in *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, vol. 34, 2021.
 - [54] F. Petroni *et al.*, “KILT: A benchmark for knowledge intensive language tasks,” in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2021, pp. 2523–2544. DOI: 10.18653/v1/2021.naacl-main.200.
 - [55] C.-Y. Lin, “ROUGE: A package for automatic evaluation of summaries,” in *Text Summarization Branches Out*, Association for Computational Linguistics, 2004, pp. 74–81. [Online]. Available: <https://aclanthology.org/W04-1013/>.
 - [56] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, “On faithfulness and factuality in abstractive summarization,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 1906–1919. DOI: 10.18653/v1/2020.acl-main.173.
 - [57] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, “RAGAs: Automated evaluation of retrieval augmented generation,” in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, 2024, pp. 150–158. DOI: 10.18653/v1/2024.eacl-demo.16.
 - [58] J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia, “ARES: An automated evaluation framework for retrieval-augmented generation systems,” in *Proceedings of the 2024 Conference of the North American Chapter of the Association*

- for *Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024, pp. 338–354. DOI: [10.18653/v1/2024.naacl-long.20](https://doi.org/10.18653/v1/2024.naacl-long.20).
- [59] L. Zheng *et al.*, “Judging LLM-as-a-judge with MT-Bench and chatbot arena,” in *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, vol. 36, 2023.
 - [60] O. Sainz, J. Campos, I. García-Ferrero, J. Etxaniz, O. L. de Lacalle, and E. Agirre, “NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark,” in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023, pp. 10 776–10 787. DOI: [10.18653/v1/2023.findings-emnlp.722](https://doi.org/10.18653/v1/2023.findings-emnlp.722).
 - [61] J. Cohen, “A coefficient of agreement for nominal scales,” *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960. DOI: [10.1177/001316446002000104](https://doi.org/10.1177/001316446002000104).
 - [62] J. L. Fleiss, “Measuring nominal scale agreement among many raters,” *Psychological Bulletin*, vol. 76, no. 5, pp. 378–382, 1971. DOI: [10.1037/h0031619](https://doi.org/10.1037/h0031619).
 - [63] J. R. Landis and G. G. Koch, “The measurement of observer agreement for categorical data,” *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977. DOI: [10.2307/2529310](https://doi.org/10.2307/2529310).
 - [64] E. B. Wilson, “Probable inference, the law of succession, and statistical inference,” *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927. DOI: [10.1080/01621459.1927.10502953](https://doi.org/10.1080/01621459.1927.10502953).
 - [65] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, UK: Cambridge University Press, 2008.
 - [66] K. L. Gwet, *Handbook of Inter-Rater Reliability*, 4th ed. Gaithersburg, MD, USA: Advanced Analytics, LLC, 2014.
 - [67] OpenAI. “Structured outputs.” API documentation. (2024), [Online]. Available: <https://platform.openai.com/docs/guides/structured-outputs> (visited on 08/30/2026).

Appendix A

Reproducibility, Environment and Cost

A.1 Execution Environments

The work was not carried out in a single environment, and this appendix records the three that were used rather than presenting a tidy composite. Each experiment directory contains a `manifest.json` recording the interpreter and library versions actually present when that experiment ran, so any run can be attributed to its environment without relying on Table A.1.

Table A.1 – Environments used, as recorded in the experiment manifests.

Env.	Python	OpenAI client	Used for
A	3.10.11	2.24.0	Candidate retrieval runs, both candidate dense-embedding runs, and the held-out generation run
B	3.10.16 (Anaconda)	2.53.0	Reviewed retrieval runs, both reviewed dense-embedding runs, and the development generation runs
C	3.10.12 (Linux)	—	Context assembly, escalation calibration, and the re-scoring of the historical baselines. No hosted call is made by any of these.

Table A.2 details environment B, the machine on which most of the work was performed.

A difference between the two generation runs. Development generation was performed in environment B and held-out generation in environment A, so the two runs used different client library versions (2.53.0 and 2.24.0) and different Python patch releases. This was not intended and is recorded because Chapter 5 compares those two runs directly. The request is determined by the prompt, the schema, the model identifier and the temperature, all of which are pinned and hashed identically in both manifests, and the model itself (`gpt-4o-mini-2024-07-18`) executes server-side; a client-version effect is therefore unlikely. It is not excluded. The runs were not repeated to test it:

Table A.2 – Hardware and principal library versions for environment B, the machine on which most of the work was performed.

Component	Value
Operating system	Windows 10.0.26200
Processor	Intel64 Family 6 Model 186 Stepping 2
Logical CPUs	16
GPU	NVIDIA GeForce RTX 4070 Laptop
Python	3.10.16 (CPython, Anaconda)
<code>openai</code>	2.53.0
<code>numpy</code>	2.2.6
<code>torch</code>	2.7.1+cu118
<code>transformers</code>	5.12.1
<code>matplotlib</code>	3.10.0
<code>python-dotenv</code>	1.0.1
<code>scikit-learn</code>	1.6.0
<code>peft</code>	0.12.0
<code>accelerate</code>	1.14.0
<code>bitsandbytes</code>	0.49.2
<code>safetensors</code>	0.5.2

responses are cached by model, prompt, schema and context, so a repeat would return the stored answers without contacting the API, and bypassing the cache would spend the held-out partition a second time, which decision D043 forbids.

A.2 Determinism and Caching

Generation used temperature 0 throughout. Every response is cached on disk under a key covering the model, the system instruction, the user message and the schema digest, so a repeated run reproduces the stored answers exactly and cannot be re-billed. The corpus embeddings were copied between cache directories rather than recomputed when the benchmark was frozen, because the corpus text did not change; the copies are recorded in the usage tables with `billed = no` so the saving is visible rather than silent.

The evaluation split is deterministic and derived from a string seed consumed by SHA-256 rather than from a random number generator. The frozen split is *inherited* from the candidate split rather than recomputed, for the reason given in decision D028.

A.3 Hosted API Usage

Two model families were used: `gpt-4o-mini-2024-07-18` for all generation, and `text-embedding-3-small` and `text-embedding-3-large` for dense retrieval. The LoRA-adapted system was executed locally on the GPU listed above and made no

hosted call.

Table A.3 reports the completion requests and their tokens.

Table A.3 – Completion requests and tokens, from the provider’s usage export.
Consolidated in docs/billing/.

Date	Run	Requests	Input	Output
2026-03-18	Few-shot baseline	300	168,466	18,913
2026-05-26	RAG baseline	300	267,952	17,268
2026-08-09	E5 generation, development	84	135,852	28,065
2026-08-12	E5 generation, held-out	56	90,638	16,462
Total		740	662,908	80,708

Table A.4 reports embedding tokens separately, deduplicated on the text digest so that copied caches are not counted twice.

Table A.4 – Embedding tokens, from the stored cache metadata, deduplicated on the text digest.

Model	Texts	Tokens	Billed
text-embedding-3-large	1,535 corpus children	171,186	yes
text-embedding-3-small	1,535 corpus children	171,186	yes
text-embedding-3-large	99 candidate questions	1,635	yes
text-embedding-3-small	99 candidate questions	1,635	yes
text-embedding-3-large	98 frozen questions	1,736	yes
text-embedding-3-small	98 frozen questions	1,736	yes
text-embedding-3-small	127 legacy chunks + 98 questions	58,762	yes
Total billed		407,876	
Reused from cache, not re-billed		342,372	no

A.4 Cost

Table A.5 lists the recorded charges.

The entire project consumed roughly twenty-nine United States cents of hosted inference. The first six charges are taken from the provider’s billing record. The two of 30 August are computed from the token counts the runs themselves report, at the same unit prices; they were incurred after the last billing export used here and are marked as such rather than presented as reconciled.

As a consistency check, five of the first six charges reconstruct exactly from the recorded token counts at unit prices of \$0.15 and \$0.60 per million tokens for input and output on `gpt-4o-mini`, and \$0.02 and \$0.13 per million input tokens for the small and large embedding models. The remaining charge, for the RAG baseline on 26 May,

Table A.5 – Recorded charges, from the provider’s cost export.

Date	What	USD
2026-03-18	Few-shot baseline generation	0.0366
2026-05-26	RAG baseline generation	0.0506
2026-07-20	Corpus and question embeddings, both models	0.0259
2026-08-08	Question re-embedding after the freeze	0.0003
2026-08-09	E5 generation, development	0.0372
2026-08-12	E5 generation, held-out	0.0235
2026-08-30	E5 generation, enriched context, development	0.0276
2026-08-30	Few-shot and RAG baselines regenerated over 98 records	0.0860
Total		0.2877

exceeds its reconstruction by \$0.00008686; that script also embeds each query before retrieving, and those embedding calls do not appear in the completions export.

The cost is worth stating plainly because it bears on the argument of this work. The expensive parts of building a defensible system here were the corpus repair, the benchmark construction, and the independent review — none of which is inference. Nothing in Chapter 5 would have been improved by spending more on model calls, and the two most useful findings, the citation-legibility defect and the reviewers’ disagreement, cost nothing at all because they came from asking practitioners rather than from running a model.

A.5 Reproducing the Results

Repository. <https://github.com/MoSaleh47/Where-RAG-Breaks---Citation-Grounded-Quest>

The repository carries the corpus builders, the retrieval, context and generation drivers, the citation evaluator, the verification layer, the test suite, the decision log and the generated-artefact log. It does not carry the reviewer identity files: the mapping from the pseudonyms R1–R4 to named people is held outside version control and is excluded by `.gitignore`, so it has never entered the repository’s history. Provider billing exports and credentials are excluded on the same basis.

Every experiment writes an immutable directory containing its outputs, a `manifest.json` with SHA-256 digests of all inputs and outputs, and the software versions in use. The order of reconstruction is:

1. `build_parent_child_corpus.py` — deterministic corpus; the rebuild is byte-identical.
2. `map_legacy_sources.py`, `build_legacy100_ai_audit.py` — lineage from the historical flat chunks and the evidence-linked candidate benchmark.

3. `freeze_reviewed_benchmark.py` with `-inherit-splits` — the frozen benchmark. Refuses incomplete or malformed review input.
4. `rerun_reviewed_retrieval.py` — all five retrieval configurations; resumes from completed stages and reuses cached vectors.
5. `rescore_legacy_baselines.py` — historical baselines under the corrected evaluator. Makes no model call.
6. `build_candidate_contexts.py`, `run_generation.py` — evidence contexts and generation. The latter defaults to the development partition and refuses the held-out partition without an explicit flag.
7. `calibrate_escalation.py` — escalation signals and rules. Fits nothing; makes no model call.
8. `capture_environment.py` — regenerates the tables in this appendix from the manifests and cache metadata.

Unit tests covering the corpus parser, the lineage mapping, grouped splitting, BM25, fusion, reranking, context assembly, the review contract, citation resolution and answer verification are run with `python -m pytest tests/`.

Appendix B

Generative AI Use Disclosure

Generative AI tools were used in this project. This appendix discloses each use in accordance with the aivancity rules: the tool, the period, the task, what was produced, and what the author verified before inclusion. The source of record is `docs/generated_artifacts_log.md` in the project repository, an append-only log of every AI-assisted artefact with its verification status; this appendix summarises it and does not replace it.

B.1 Three distinct kinds of use

It is important to separate three things that are all, loosely, “AI use”, because they carry different obligations.

1. Models under study. `gpt-4o-mini-2024-07-18` is the generator whose behaviour Chapter 5 measures; the two `text-embedding-3` models are components of the retrieval system being evaluated. These are the object of the research, not tools used to write it. Every call is recorded in an experiment manifest with its token count, and the full usage and cost record is in Appendix A. The historical benchmark questions were themselves generated by a hosted model in the earlier phase of the project; this is why the benchmark is described throughout as synthetic in origin and why the independent review in Section 5.2.2 was run.

2. Development assistance. OpenAI Codex was used in July 2026 for repository inspection, test and script implementation, experiment-manifest reconciliation, and LaTeX conversion. Its output was independently re-verified before acceptance (artefact log, 25 July: “Independent verification of the reconciliation”), and that verification found one typesetting regression and one unconfirmed test count, both recorded.

3. Research and writing assistance. Claude (Anthropic), accessed through the Claude desktop application, was used from July to August 2026 as an assistant for analysis, code, and drafting. This is the most substantial use and is itemised below.

B.2 Itemised record of research and writing assistance

Table B.1 itemises each AI-assisted artefact, the task it assisted with, and what was verified before it entered this document.

Table B.1 – AI-assisted artefacts, from the artefact log. “Verified” states what the author checked before the artefact was accepted.

Date	Produced with AI assistance	Verified by the author
25 Jul	Review-form generator, response aggregator, pseudonymisation script	Round-tripped against synthetic responses; a defect in the κ computation was found and fixed
25 Jul	Chapter 2 draft; bibliography from 10 to 64 entries	Every entry checked against a primary record; the chapter marked pending author edit
25 Jul	Chapter 4 draft	Every quantity traced to a manifest; eight repository discrepancies recorded rather than smoothed
27 Jul	Chapter 1 and 3 drafts	Regulatory content read from the validated corpus, no provision asserted from memory
8 Aug	Adjudication tool; freeze; retrieval re-scoring; corrected citation evaluator with 29 tests	Contract validated; a split-recomputation defect caught and reversed (D028); the old scorer’s errors pinned by tests
8 Aug	Generation runner; verification rule; prompt v2; escalation calibration	Held-out split protected by tooling; every result compared with its artefact; one prose claim corrected by the artefact (see log)
8–15 Aug	Chapters 5 and 6, abstract, appendices; consistency corrections to Chapters 1–4	Every figure read from a stored metrics file; the two properties no internal metric detects checked independently

B.3 What was not delegated

The following were decided by the author and not by any AI system.

- The research question and its reframing from a system comparison into a failure analysis.
- Every one of the 44 recorded decisions in the decision log, including the choice to run an independent review, the pre-registered cutoff, the choice of generator, and the decision not to fit an escalation threshold.
- The adjudication of the 19 contested benchmark records, each with a recorded justification.
- The authorisation of every hosted API call.

- Whether each AI-proposed correction was accepted. Three proposed adjudication repairs (D026) were approved by the author before application, with the pre-repair file retained.

B.4 Provenance discipline

Two rules governed AI-assisted work throughout. Every number in this document comes from a stored, hashed artefact rather than from prose, so an AI-drafted sentence cannot introduce a figure that does not exist in the record. And where the artefact and the prose disagreed, the artefact was allowed to overrule the prose; the artefact log records the occasions on which that happened.

The chapters were drafted with assistance and are the author's responsibility in full. The author has read, and where necessary rewritten, every chapter, and is prepared to defend every claim in them orally.

Appendix C

Independent Benchmark Review Protocol

The independent review covers 33 unique question records: all 22 audit exceptions, all three hybrid top-20 residuals, and ten deterministic unchanged controls stratified by source class and partition.

For each record, the reviewer assesses:

- whether the question is clear and unambiguous;
- whether every material answer claim is supported;
- whether the canonical citation is correct;
- whether the selected evidence is complete;
- whether alternative one-child or multi-child evidence routes should be accepted;
- whether the question, answer, citation, parent, or evidence identifiers require correction;
- whether the record should be excluded or marked uncertain.

The reviewer records a name or role and an ISO-format review date. The freeze process rejects missing rows, duplicated identifiers, unknown evidence, incomplete decision fields, and mismatched source hashes. Reviewed exceptions and controls are distinguished from unselected candidate records; the report does not imply that every benchmark row received expert legal review.

Appendix D

Selected Source Code

The full codebase is in the repository given in Appendix A. This appendix reproduces the four excerpts on which the claims of Chapters 4 and 5 most directly depend, so that a reader can check the argument against the implementation without cloning anything. Each listing is quoted verbatim from the file and line range named in its caption; nothing is paraphrased for presentation.

D.1 Quote Grounding

Listing D.1 is the check the central claim of this work rests on. `quote_is_grounded` accepts a quote only when its word sequence appears *contiguously* in the cited evidence, after typographic and punctuation normalisation. Contiguity is what rejects a quote assembled from two fragments, which is the failure a syntactic search for ellipsis characters passes (Section 5.5). The byte-level rule is kept alongside it and reported as a second figure throughout, rather than replaced.

```
1 def _normalize(value: str) -> str:
2     """Whitespace and case only -- the original, strictest rule."""
3     return WHITESPACE.sub(" ", value).strip().casefold()
4
5
6 def _normalize_typography(value: str) -> str:
7     value = unicodedata.normalize("NFKC", value)
8     for source, target in _TYPOGRAPHY.items():
9         value = value.replace(source, target)
10    return value
11
12
13 def _normalize_for_matching(value: str) -> str:
14     """Typography- and punctuation-insensitive; word order is
15        preserved."""
16    value = NON_WORD.sub(" ", _normalize_typography(value))
17    return WHITESPACE.sub(" ", value).strip().casefold()
18
```

```

19 def quote_is_grounded(quote: str, evidence_text: str) -> bool:
20     """Does the quote's word sequence appear contiguously in the
        evidence?"""
21     return _normalize_for_matching(quote) in
        _normalize_for_matching(evidence_text)
22
23
24 def quote_is_verbatim(quote: str, evidence_text: str) -> bool:
25     """Byte-level check under the original whitespace-and-case
        rule."""
26     return _normalize(quote) in _normalize(evidence_text)
27
28
29 def contains_elision(quote: str) -> bool:
30     """Does the quote mark omitted material with an ellipsis?"""
31     return bool(ELLIPSIS.search(quote))

```

Listing D.1 – The contiguity definition of a verbatim quote, with the byte-level rule retained beside it. Source:

src/aviation_rag/generation/verification.py, lines 60–90.

D.2 Structural Verification

Listing D.2 is the loop that decides whether a claim is supported. Two properties are worth reading off the code directly. A support whose `source_id` is absent from `evidence_by_id` fails immediately — this is the check that caught the fabricated identifier `EU_376_2014_ART_5__C001` on `QA_0017`, inside a response the provider's strict structured-output mode had already accepted. And a claim can lose its support without contributing to `quote_errors`, which is why Section 5.5 reads support coverage from the list of supported claim identifiers rather than inferring it from the decision.

```

1         claim_id = claim.get("claim_id", "?")
2         claim_supported = True
3         for support in claim.get("supports", []):
4             if not isinstance(support, dict):
5                 claim_supported = False
6                 continue
7             source_id = support.get("source_id")
8             quote = support.get("quote")
9             if (
10                 source_id not in evidence_by_id
11                 or not isinstance(quote, str)
12                 or not quote.strip()
13             ):
14                 claim_supported = False

```

```

15         continue
16         evidence_text = evidence_by_id[source_id]["text"]
17         if contains_elision(quote):
18             elided_quotes.append(f"{claim_id}: elided
19                                   quote from {source_id}")
20         if not quote_is_grounded(quote, evidence_text):
21             quote_errors.append(
22                 f"{claim_id}: quote not found in
23                   {source_id}"
24             )
25             claim_supported = False
26         if not quote_is_verbatim(quote, evidence_text):
27             strict_errors.append(
28                 f"{claim_id}: quote not byte-verbatim in
29                   {source_id}"
30             )
31         if claim_supported and claim.get("claim_id"):

```

Listing D.2 – The support loop: supplied-source admissibility, contiguity, and the byte-level second opinion. Source: `src/aviation_rag/generation/verification.py`, lines 105–132.

D.3 Citation Scoring

The evaluator that replaces the historical scorer resolves references against the corpus instead of matching strings. Listing D.3 is the exact measure, and the reason 10 of the 98 frozen records fall outside its denominator rather than scoring zero: a gold citation that no parser can resolve is usually one no reader can resolve either, and that count is reported as a finding in Section 5.7.

```

1 def exact_citation_match(prediction: str, gold_citation: str) ->
2   tuple[bool | None, list[str]]:
3     """Is every atomic unit of the gold citation named in the
4       prediction?
5
6       Returns '(None, [])' when the gold citation contains no
7       parseable unit --
8       for instance 'GM to Reg. (EU) No 376/2014 [04]' or a bare
9       'Paragraph 3'.
10      Those records are excluded from the exact metric's denominator
11      rather than
12      silently scored zero, and their count is itself a finding: a
13      citation label
14      a parser cannot resolve is usually one a reader cannot resolve
15      either.

```

```

9      """
10     gold_units = parse_citations(gold_citation)
11     if not gold_units:
12         return None, []
13     predicted = {citation.signature() for citation in
14                  parse_citations(prediction)}
15     missing = [str(unit) for unit in gold_units if
16               unit.signature() not in predicted]
17     return (not missing), missing

```

Listing D.3 – Exact citation matching, with unresolvable gold labels excluded rather than failed. Source: `src/aviation_rag/evaluation/citations.py`, lines 314–328.

Listing D.4 handles the ambiguity the benchmark itself carries. Where a gold citation names a provision without naming which of the four regulations it belongs to, every regulation holding that provision counts as a candidate. A prediction is therefore never marked wrong for failing to draw a distinction the gold label does not draw.

```

1  def resolve_candidates(self, citation: Citation) -> set[str]:
2      """Every parent the citation could denote, given what it
3         actually says.
4
5         A citation that names its regulation denotes exactly one
6         parent. A bare
7         ‘Article 2’ denotes any of the four regulations that
8         have an Article 2,
9         and this returns all of them. Scoring against the
10        candidate set rather
11        than against the default attribution means a prediction is
12        never marked
13        wrong for failing to make a distinction that the gold
14        citation itself
15        does not make.
16        """
17
18     if not citation.defaulted_regulation:
19         parent = self.resolve(citation)
20         return {parent} if parent else set()
21     candidates = set()
22     for regulation in REGULATION_PREFIX:
23         parent = self.resolve(
24             Citation(regulation, citation.kind,
25                     citation.number, citation.subs)
26         )
27         if parent:
28             candidates.add(parent)
29     return candidates

```

Listing D.4 – Ambiguous gold citations resolve to a candidate set, not to a default attribution. Source: `src/aviation_rag/evaluation/citations.py`, lines 138–158.